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# Olist E-commerce DWH Architecture & Analytics

## From Data to Business Insights

Building an Enterprise Data Warehouse and extracting high-value operational strategy using Advanced SQL.

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**SQL Server | T-SQL | SSIS ETL | Data Warehouse | Business Analytics**

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# Building the Data Warehouse from Scratch

I designed and built the warehouse end-to-end, starting from raw source data and moving through the layered architecture.

## Source Systems

- CRM
- ERP
- CSV Files

→

## Bronze Layer

- Raw data
- Minimal changes
- Data landing

→

## Silver Layer

- Cleaned data
- Standardized
- Business rules

→

## Gold Layer

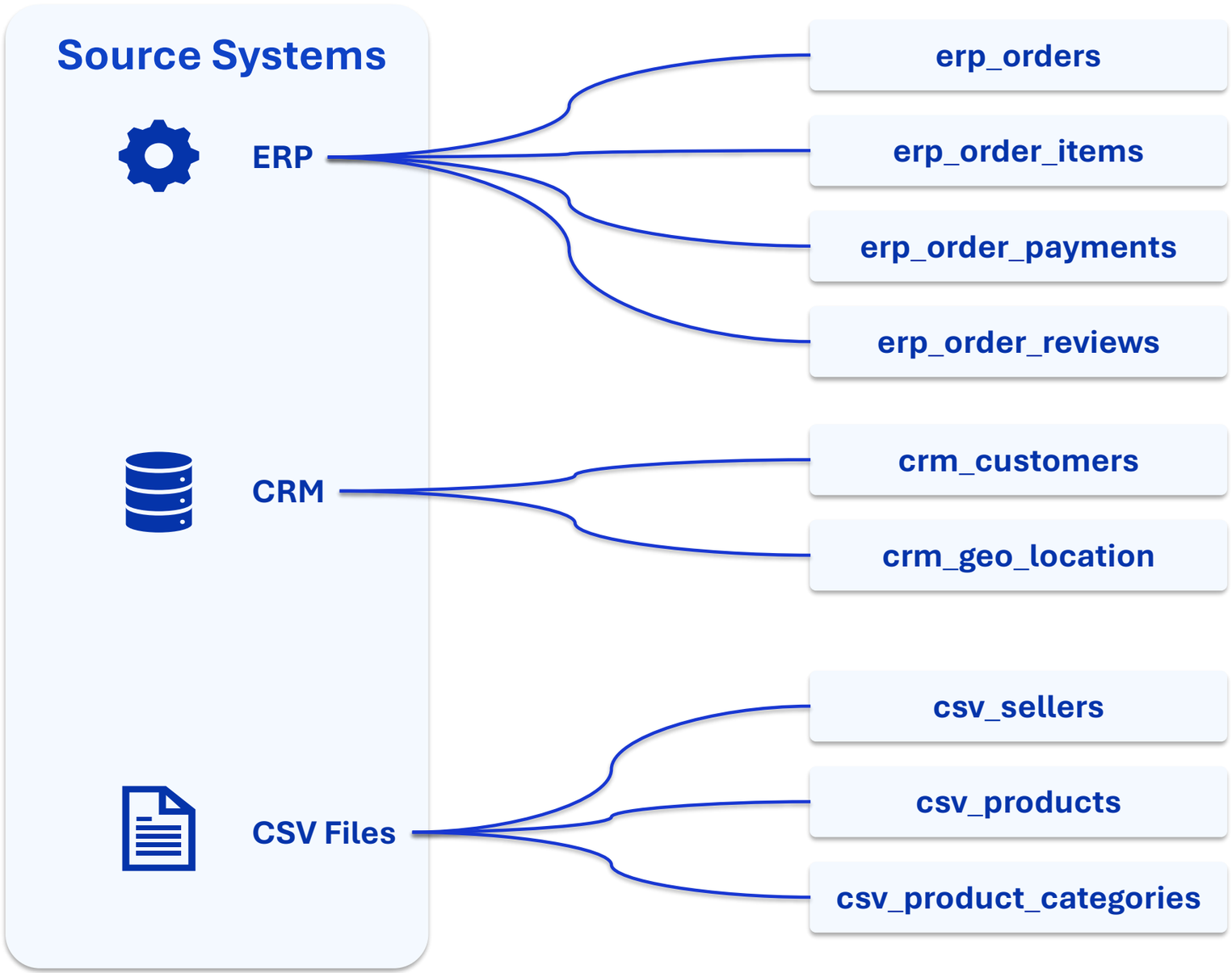
- Optimized data
- Ready for analysis
- Galaxy schema

**SSIS ETL Pipeline Orchestration**

Automated data extraction, transformations, data validation, and incremental loading between layers.

# 1- Source Systems & Data Ingestion :

The project integrates data from multiple source systems to establish a centralized foundation for the data warehouse.



## 2- Bronze Layer: Raw Data Ingestion

The Bronze Layer serves as the initial landing zone for data ingested from source systems.

### Bronze Layer



- Raw data
- Minimal changes
- Data landing

### Key Responsibilities

Load source data into the warehouse environment.

### Raw Data Storage

Preserve ingested records before downstream transformations.

### Data Traceability

Maintain a clear starting point for subsequent processing stages.

# 3- Silver Layer: Data Cleansing & Transformation

The Silver Layer transforms raw data into cleaner, standardized, and more reliable datasets using SQL.

**Silver Layer**



- Cleaned data
- Standardized
- Business rules

| | |
|---|---|
| **Data Cleaning** | Handle duplicates and investigate data quality issues. |
| **Standardization** | Align data types and formats. |
| **Data Validation** | Compare records and investigate inconsistencies across processing layers. |
| **Business Rules** | Apply transformation logic before loading the Gold Layer. |
| **Key Objective** | Improve data consistency and reliability before preparing the data for analytical modeling. |

# Silver Layer

## A- Key Data Quality Interventions

### Temporal Alignment

2-stage CTE logic to correct impossible order timestamp sequences automatically.

### Window Deduplication

ROW_NUMBER partitioning to isolate latest order reviews and purge duplicates.

### Geographic Cleansing

State code translation and substring parsing to fix dirty city names and emails.

### Null Imputation

COALESCE fallbacks preventing null propagation across product dimensions.

## B- Geographic String Standardizations

### State Mapping

Mapping 27 Brazilian 2-letter state abbreviations to canonical full names (e.g., 'SP' ➔ 'Sao Paulo').

### Abbreviation Fixes

Replacing shorthand city tokens like 'sbc' with full name 'sao bernardo do campo'.

### Delimiter Parsing

Using CHARINDEX to slice trailing artifacts separated by slashes ('/'), dashes ('-'), or brackets.

## C- Geographic String Standardizations

### String Normalization

**Trim Quotes:** Strip leading/trailing quotation marks from identifiers.

**Uppercase IDs:** Standardize hash keys to uppercase for join integrity.

**Text Standard:** Replace underscores with spaces in category titles.

### Numerical Safeguards

**Zero Imputation:** COALESCE missing product dimensions & weight to 0.

**Coordinate Rounding:** Truncate lat/lng floats to 6 decimal precision.

**Default Strings:** Assign 'n/a' to missing review comments & titles.

# Silver Layer

## Multi-Stage Temporal Sanitization

**The Ingestion Anomaly**

Operational logs recorded carrier pickup timestamps prior to purchase or approval dates.

```
with cleaned_a as (
    SELECT
       UPPER(REPLACE(TRIM(order_id), '"','')) as order_id,
       UPPER(REPLACE(TRIM(customer_id), '"','')) as customer_id,
       UPPER(TRIM(order_status)) as order_status,

       Case
          when order_purchase_timestamp > order_delivered_carrier_date
               and order_approved_at  > order_delivered_carrier_date
            then order_delivered_carrier_date
          else order_purchase_timestamp
       end as order_purchase_timestamp,

       Case
          when order_purchase_timestamp > order_delivered_carrier_date
               and order_approved_at  > order_delivered_carrier_date
            then order_purchase_timestamp
          when order_approved_at  > order_delivered_carrier_date
               and order_approved_at  > order_delivered_customer_date
            and order_approved_at  > order_estimated_delivery_date
            then order_delivered_carrier_date
          when order_approved_at  > order_delivered_carrier_date
               and order_approved_at  > order_delivered_customer_date
            then order_delivered_carrier_date
          when order_approved_at  > order_delivered_carrier_date
            then order_delivered_carrier_date
          when order_approved_at  > order_estimated_delivery_date
            then order_estimated_delivery_date
          else order_approved_at
       end as order_approved_at,
```

```
       Case
         when order_purchase_timestamp > order_delivered_carrier_date
              and order_approved_at  > order_delivered_carrier_date
           then order_approved_at
         when order_approved_at  > order_delivered_carrier_date
              and order_approved_at  > order_delivered_customer_date
              and order_approved_at  > order_estimated_delivery_date
           then order_delivered_customer_date
         when order_approved_at  > order_delivered_carrier_date
              and order_approved_at  > order_delivered_customer_date
           then order_delivered_customer_date
         when order_approved_at  > order_delivered_carrier_date
           then order_approved_at
         when order_delivered_carrier_date  > order_delivered_customer_date
           then order_delivered_customer_date
         when order_delivered_carrier_date  > order_estimated_delivery_date
           then order_estimated_delivery_date
         else order_delivered_carrier_date
       end as order_delivered_carrier_date,

       case
         when order_approved_at  > order_delivered_carrier_date
              and order_approved_at  > order_delivered_customer_date
              and order_approved_at  > order_estimated_delivery_date
           then order_estimated_delivery_date
         when order_approved_at  > order_delivered_carrier_date
              and order_approved_at  > order_delivered_customer_date
           then order_approved_at
         when order_delivered_carrier_date  > order_delivered_customer_date
           then order_delivered_carrier_date
         else order_delivered_customer_date
       end as order_delivered_customer_date,
```

```
       case
         when order_approved_at  > order_delivered_carrier_date
              and order_approved_at  > order_delivered_customer_date
              and order_approved_at  > order_estimated_delivery_date
           then order_approved_at
         when order_approved_at  > order_estimated_delivery_date
           then order_approved_at
         when order_delivered_carrier_date  > order_estimated_delivery_date
           then order_delivered_carrier_date
         else order_estimated_delivery_date
       end as order_estimated_delivery_date
    FROM olist_dwh.bronze.erp_Orders),
```

# Silver Layer

## Window-Based Review Duplicates

### On Review per Order

Retains strictly rn = 1, isolating the single most recent feedback submission per purchase.

```
FROM
(
    SELECT
        UPPER(REPLACE(TRIM(review_id), '"', '')) AS review_id,
        UPPER(REPLACE(TRIM(order_id), '"', '')) AS order_id,
        review_score,
        COALESCE(review_comment_title, 'n/a') AS review_comment_title,
        COALESCE(review_comment_message, 'n/a') AS review_comment_message,
        review_creation_date,
        review_answer_timestamp,

        ROW_NUMBER() OVER
        (
            PARTITION BY
                UPPER(REPLACE(TRIM(order_id), '"', ''))
            ORDER BY
                review_creation_date DESC,
                review_id
        ) AS rn

    FROM olist_dwh.bronze.erp_order_reviews
) AS ranked_reviews
WHERE rn = 1;
```

## 4- Gold Layer: Dimensional Modeling

The Gold Layer organizes transformed data into a dimensional model designed for business analysis and reporting.

### Gold Layer

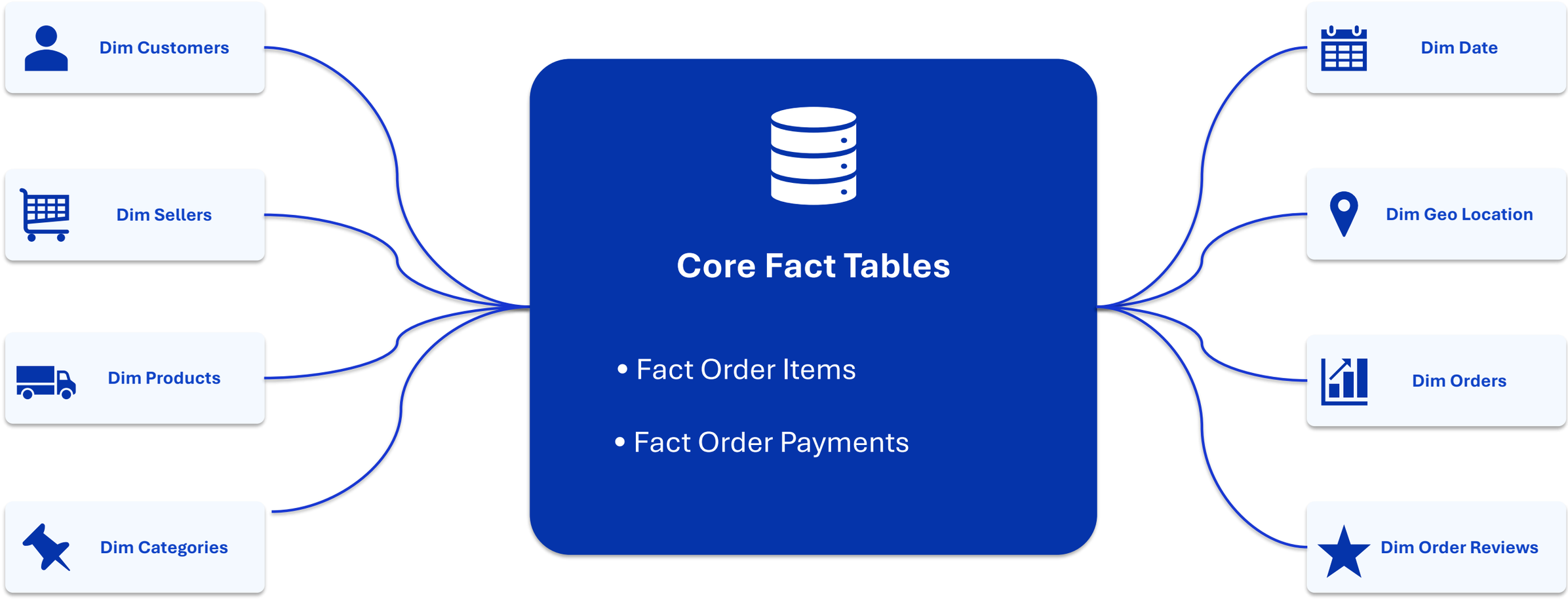


- Optimized data
- Ready for analysis
- Galaxy schema

| | |
|---|---|
| **Dimension Tables** | Provide descriptive business context. |
| **Fact Tables** | Store measurable business events. |
| **Galaxy Schema** | Support analysis across related business processes. |
| **Analytics-Ready Data** | Enable SQL-based analysis and downstream BI reporting. |
| **Key Objective** | Deliver a structured analytical data model that supports business questions and data-driven decision-making. |

# Data Modeling (Galaxy Schema Architecture)

I designed a dimensional model using a Galaxy Schema to support flexible analysis across multiple business domains.



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PHASE 01

# Exploratory Data Analysis (EDA)

EDA — DATASET OVERVIEW

# Data Volume & Metric Scale (2016–2018)

| | | | |
|---|---|---|---|
| **99.44k** Total Orders | **16.01M** Total Revenue | **161** AOV | **112.65K** Total Qty Sold |
| **3.1K** Sellers | **96. 1K** Customers | **32.95K** Total Products | **71** Total Categories |
| **27** States | **6K** Cities | **12.5** Avg Delivery Days | |

**Timeline Scope: September 2016 through October 2018 (Peak operational growth in 2017-2018)**

EDA — DATA QUALITY & FULFILLMENT

# Order Status Split & Missing Data Audit

### Order Status Breakdown




### ETL Cleanse Highlights

**Delivery Dates Nulls:** 2.98% nulls present in un-delivered orders.

**Category Translation:** 1.89% Missing product category names.

**Review Comments:** 58% empty review message strings handled cleanly in DW.

**Order Items Completeness:** 775 orders have no corresponding order-item records in the Gold layer.

**Payment-to-Items Gap:** 830 payment records belong to orders without corresponding order-item records, requiring further source-data investigation.

EDA — PRICE & FREIGHT STATS

# Price, Freight Value & Delivery Baselines

### Item Price Distribution

**Average Price:** 120.65 $

**Median Price:** 74.99 $

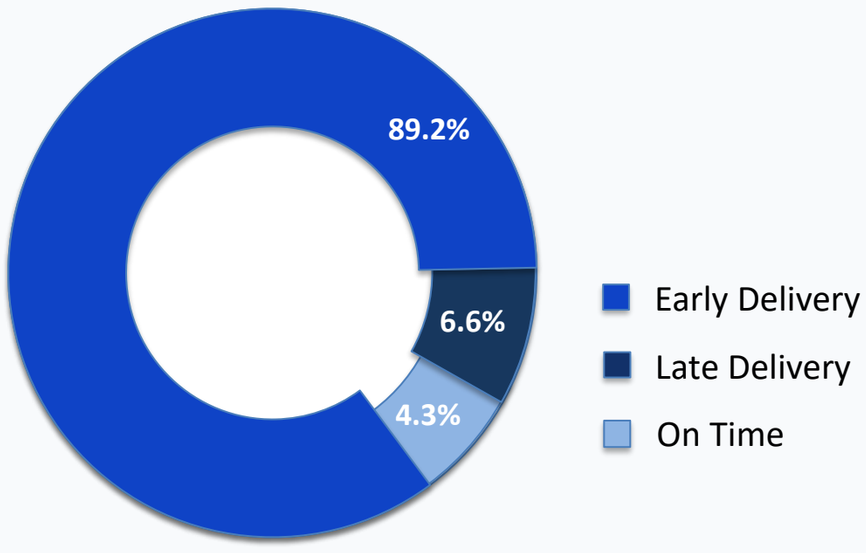
**High Skewness:** Heavy concentration under 200 $

### Freight Cost Distribution

**Average Freight:** 19.99 $

**Freight Ratio:** 16% of total order value

**Regional Variance:** North region pays 2x higher freight




### Orders by Delivery Status

**Early delivery:** Achieved **88,649** of **99,441** Orders.

**Late delivery:** **6,520** out of **99,441** Orders are Late.

**On Time:** **4,272** out of **99,441** requests have been completed.

### Delivery Time Baseline

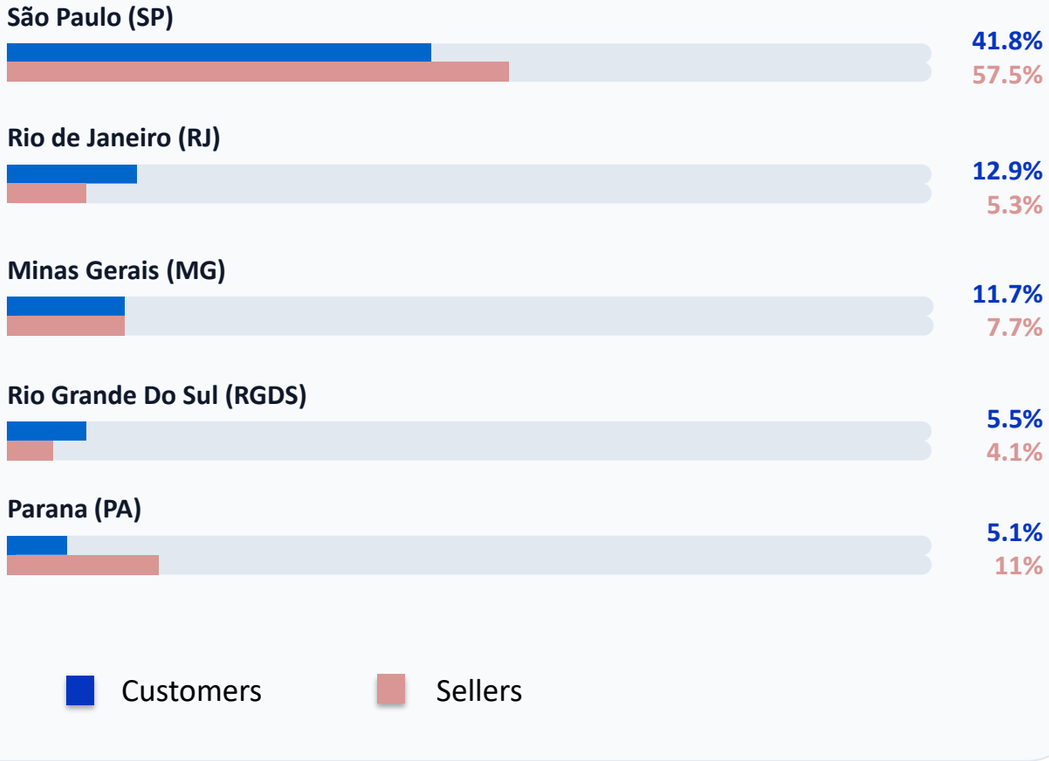
**Average Delivery:** 12.5 days

**Estimated Lead:** 23.2 days target

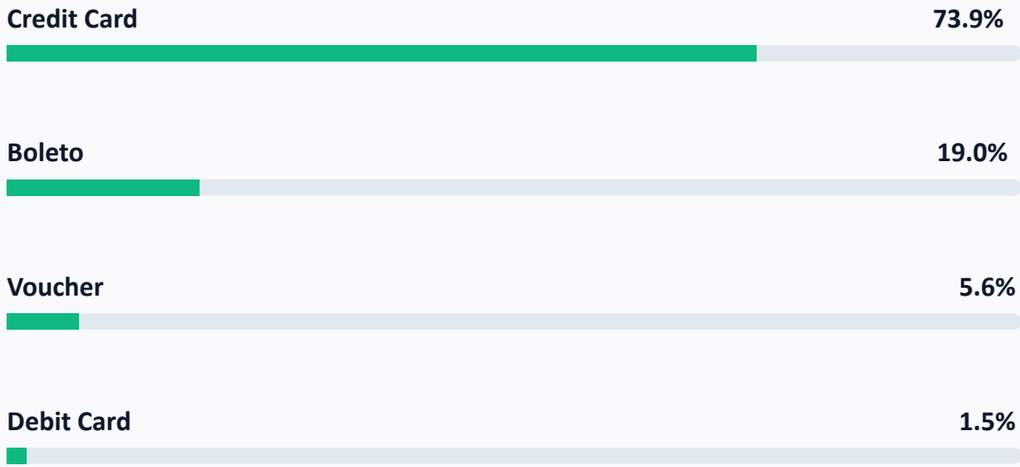
EDA — GEOGRAPHY & PAYMENTS

# Geographic Concentration & Payment Mix

### Top States by Customer Volume, and Sellers




### Payment Method Distribution



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## PHASE 02

# Business Questions & Advanced SQL

# Analysis Approach

To answer the business questions, I used the Gold Layer and applied:

- CTEs
- COUNT(DISTINCT)
- Window Functions
- Joins across Facts & Dimensions
- Time-based Analysis
- Aggregations
- CASE WHEN
- Date Functions
- Customer/Seller Segmentation

**The result: clean, reliable, and well-structured data ready for business analysis.**

# Business Questions & Analysis Topics

I analyzed the following areas to uncover key business insights:

- Customers
- Sellers
- Products
- Orders
- Sales / Revenue
- Geographic Analysis
- Payments
- Reviews

# Business Questions



TOPIC 01 OF 08

## 1. Customers Analytics

New vs Repeating Customers, Customer Value, Retention Patterns, Customers Segmentation, Purchase Behavior

**Proceed to Analytical Questions & Queries** ➡

## Customer Analysis — Business Question

**Question - 1** What percentage of customers are repeat customers vs. one-time buyers?

### SQL Query

1

```sql
With Customer_Orders As (
    select
        c.customer_unique_id,
        count(o.order_sk) As total_orders,
        Case
        When count(o.order_sk) = 1 Then 'One-Time Customer'
        Else 'Repeat Customer'
    End As Customer_Type
    from olist_dwh.gold.dim_orders o
    join olist_dwh.gold.dim_customers c
        on o.customer_sk = c.customer_sk
    group by c.customer_unique_id
),
Customer_Revenue As (
    select
        c.customer_unique_id,
        Sum(op.payment_value) As Total_Revenue
    from olist_dwh.gold.dim_orders o
    join olist_dwh.gold.dim_customers c
        on o.customer_sk = c.customer_sk
    join olist_dwh.gold.fact_order_payments op
        on o.order_sk = op.order_sk
    group by c.customer_unique_id
)
```

2

```sql
select
    co.Customer_Type,
    Count(co.customer_unique_id) As Customer_Count,

    Concat(
        Cast(
            Count(co.customer_unique_id) * 100.00
            / Sum(Count(co.customer_unique_id)) over()
              As decimal(5,2))
            , '%'
    ) As Percentage_of_Customers,
    Sum(cr.Total_Revenue) As Total_Revenue,

    Concat(
        Cast(
            (Sum(cr.Total_Revenue)
            / Sum(Sum(cr.Total_Revenue)) over()) * 100
              As decimal(5,2))
            , '%'
    ) As Percentage_of_Revenue,

    Cast(
        Sum(cr.Total_Revenue)
            / Count(co.customer_unique_id)
              As decimal(5,2)
    ) As Revenue_per_customer

From Customer_Orders co
join Customer_Revenue cr
    on co.customer_unique_id = cr.customer_unique_id
group by Customer_Type
order by Customer_Count desc;
```

### One-Time Customer

**92379 Customers || 96.86%**

- **Revenue: 14.82 M**
- **Pct of Revenue: 93.51%**
- **Revenue per Customer: 160.38**

### Repeat Customer

**2994 Customers || 3.14%**

- **Revenue: 1.02 M**
- **Pct of Revenue: 6.49%**
- **Revenue per Customer: 343.29**

## Customer Analysis — Business Question

**Question - 2** Who are the top customers in terms of revenue, Orders, Avg sales, AOV?

**SQL Query**

```sql
select
    Top 5
    c.customer_unique_id,
    Count(Distinct o.order_sk) As Total_Orders,
    Sum(op.payment_value) As Total_Revenue,
    CAST(AVG(op.payment_value) AS decimal(10,2)) As Avg_Revenue,
    CAST(Sum(op.payment_value) / Count(Distinct o.order_sk) As decimal(10,2)) AS AOV,
    CASE
        WHEN Sum(op.payment_value) > 3000 THEN 'High-value customers'
        WHEN Sum(op.payment_value) > 1000 THEN 'Regular-value customers'
        ELSE 'low-value customers'
    END As Customer_segment
from olist_dwh.gold.dim_orders o
join olist_dwh.gold.dim_customers c
    on o.customer_sk = c.customer_sk
join olist_dwh.gold.fact_order_payments op
    on o.order_sk = op.order_sk
group by c.customer_unique_id
order by Total_Orders desc;
```

| Customer Unique Id | Total Orders | Total Revenue | Avg Revenue | AOV | Customer segment |
|---|---|---|---|---|---|
| 8D50F5EADF50201CCDCEDFB9E2AC8455 | 17 | 2819.85 | 165.87 | 165.87 | Regular-value customers |
| 3E43E6105506432C953E165FB2ACF44C | 9 | 1354.40 | 150.49 | 150.49 | Regular-value customers |
| 6469F99C1F9DFAE7733B25662E7F1782 | 7 | 972.85 | 138.98 | 138.98 | low-value customers |
| CA77025E7201E3B30C44B472FF346268 | 7 | 807.61 | 115.37 | 115.37 | low-value customers |
| 1B6C7548A2A1F9037C1FD3DDFED95F33 | 7 | 652.49 | 93.21 | 93.21 | low-value customers |

## Customer Analysis — Business Question

**Question - 3** Which customer segments generate the highest revenue, what is the AOV by customer segment?

### SQL Query

```sql
WITH Customer_Status As (
    select
        c.customer_unique_id,
        Count(Distinct o.order_sk) As Total_Orders,
        Sum(op.payment_value) As Total_Revenue,
        CAST(AVG(op.payment_value) AS decimal(10,2)) As Avg_Revenue,
        CASE
            WHEN Sum(op.payment_value) > 3000 THEN 'High-value customers'
            WHEN Sum(op.payment_value) > 1000 THEN 'Regular-value customers'
            ELSE 'low-value customers'
        END As Customer_segment
    from olist_dwh.gold.dim_orders o
    join olist_dwh.gold.dim_customers c
        on o.customer_sk = c.customer_sk
    join olist_dwh.gold.fact_order_payments op
        on o.order_sk = op.order_sk
    group by c.customer_unique_id
)
select
    Customer_segment,
    COUNT(customer_unique_id) As Total_Customers,
    CAST(
        COUNT(customer_unique_id) * 100.00
        /
        (select COUNT(*)
        from Customer_Status
        )
        As decimal(10,2)
    ) As Percentage_of_Customers,
    SUM(Total_Orders) As Total_Orders,
    SUM(Total_Revenue) As Total_Revenue,
    CAST(Sum(Total_Revenue) / SUM(Total_Orders) As decimal(10,2)) AS AOV
FROM Customer_Status
GROUP BY Customer_segment
ORDER BY Total_Revenue DESC;
```

### Low-value Customer

**94882 Customers || 98.74%**

- Orders: 98062
- Revenue: 14.08 M
- AOV: 143.58

### Regular Customer

**1160 Customers || 1.21%**

- Orders: 1321
- Revenue: 1.71 M
- AOV: 1297.05

### High Customer

**53 Customers || 0.06%**

- Orders: 57
- Revenue: 0.22 M
- AOV: 3781.66

## Customer Analysis — Business Question

**Question - 4** Are high-value customers concentrated in specific regions?

### SQL Query

```sql
WITH Customer_Status As (
    select
        c.customer_unique_id,
        l.state,
        Count(Distinct o.order_sk) As Total_Orders,
        Sum(op.payment_value) As Total_Revenue,
        CASE
            WHEN Sum(op.payment_value) > 3000 THEN 'High-value customers'
            WHEN Sum(op.payment_value) > 1000 THEN 'Regular-value customers'
            ELSE 'low-value customers'
        END As Customer_segment
    from olist_dwh.gold.dim_orders o
    join olist_dwh.gold.fact_order_payments op
        on o.order_sk = op.order_sk
    join olist_dwh.gold.dim_customers c
        on o.customer_sk = c.customer_sk
    join olist_dwh.gold.dim_geo_location l
        on c.geo_location_sk = l.geo_location_sk
    group by c.customer_unique_id, l.state
),
State_Summary As (
    select
        state,
        Count(*) As Total_Customers,
        SUM(
            CASE
               When Customer_segment = 'High-value customers' Then 1
               Else 0
            End
        ) As High_value_customers
    from Customer_Status
    group by state

)
select
    state,
    Total_Customers,
    High_value_customers,
    CAST(
        High_value_customers * 100.00 / Total_Customers
        As decimal(10,2)
    ) As High_value_customers_Rate,
    dense_rank() over(order by CAST(
        High_value_customers * 100.00 / Total_Customers
        As decimal(10,2)
    ) Desc) As Rank
from State_Summary
where High_value_customers > 0
order by High_value_customers_Rate Desc;
```

| State | Total Customers | High Value Customer | Rank |
|---|---|---|---|
| BAHIA | 3259 | 6 | 1 |
| MARANHAO | 722 | 1 | 2 |
| MATO GROSSO DO SUL | 694 | 1 | 2 |
| PARA | 945 | 1 | 3 |
| GOIAS | 1944 | 2 | 4 |
| RIO DE JANEIRO | 12371 | 9 | 5 |
| SANTA CATARINA | 3534 | 2 | 6 |
| PARANA | 4869 | 3 | 6 |
| SAO PAULO | 40283 | 23 | 7 |
| DISTRITO FEDERAL | 1904 | 1 | 7 |
| MINAS GERAIS | 11247 | 4 | 8 |

## Customer Analysis — Business Question

**Question - 5** Which States have the highest customer concentration, and represent the largest customer markets?

### SQL Query

```sql
select
    TOP 5
    Coalesce(l.state, 'N/A') As State,
    Count(Distinct customer_unique_id) As Customers,
    Concat(
        Cast(
            Count(Distinct customer_unique_id) * 100.00
            / Sum(Count(Distinct customer_unique_id)) over()
          As decimal(5,2)),'%'
    ) As Percentage_of_Customers
from olist_dwh.gold.dim_customers c
Left join olist_dwh.gold.dim_geo_location l
    On c.geo_location_sk = l.geo_location_sk
group by l.state
order by Customers desc;
```

1. **SAO PAULO** — 40284 Customers || 41.90 %
2. **RIO DE JANEIRO** — 12371 Customers || 12.87 %
3. **MINAS GERAIS** — 11247 Customers || 11.70 %
4. **RIO GRANDE DO SUL** — 5273 Customers || 5.48 %
5. **PARANA** — 4869 Customers || 5.06 %

## Customer Analysis — Business Question

**Question - 6** Which Cities have the highest customer concentration, and represent the largest customer markets?

### SQL Query

```sql
select
    TOP 5
    Coalesce(l.city, 'N/A') As city,
    Count(Distinct customer_unique_id) As Customers,
    Concat(
        Cast(
            Count(Distinct customer_unique_id) * 100.00
            / Sum(Count(Distinct customer_unique_id)) over()
          As decimal(5,2)),'%'
    ) As Percentage_of_Customers
from olist_dwh.gold.dim_customers c
Left join olist_dwh.gold.dim_geo_location l
    On c.geo_location_sk = l.geo_location_sk
group by l.city
order by Customers desc;
```

| | | |
|---|---|---|
| 1 | SAO PAULO | 14982 Customers \|\| 15.57 % |
| 2 | RIO DE JANEIRO | 6620 Customers \|\| 6.88 % |
| 3 | BELO HORIZONTE | 2672 Customers \|\| 2.78 % |
| 4 | BRASILIA | 1898 Customers \|\| 1.97 % |
| 5 | CURITIBA | 1465 Customers \|\| 1.52 % |

## Question - 7 Which State generate the highest revenue per customer?

### SQL Query

```sql
select
    Top 5
    Coalesce(l.state, 'N/A') As State,
    SUM(op.payment_value) As Total_Revenue,
    Count(Distinct c.customer_unique_id) As Customers,
    Cast(
        SUM(op.payment_value) / Count(Distinct customer_unique_id)
        As decimal(5,2)
    ) As Revenue_per_customer
from olist_dwh.gold.fact_order_payments op
JOIN olist_dwh.gold.dim_orders o
    ON op.order_sk = o.order_sk
JOIN olist_dwh.gold.dim_customers c
    ON o.customer_sk = c.customer_sk
JOIN olist_dwh.gold.dim_geo_location l
    ON c.geo_location_sk = l.geo_location_sk
group by l.state
order by Revenue_per_customer desc;
```

1. **PIAUI**

| Customers | Revenue | Revenue per Customer |
|---|---|---|
| 479 | 89.61 K | 187.08 |

2. **PARAIBA**

| Customers | Revenue | Revenue per Customer |
|---|---|---|
| 517 | 92.31 K | 178.55 |

3. **PARA**

| Customers | Revenue | Revenue per Customer |
|---|---|---|
| 945 | 167.41 K | 177.15 |

4. **UNKNOWN**

| Customers | Revenue | Revenue per Customer |
|---|---|---|
| 291 | 50.97 K | 175.15 |

5. **BAHIA**

| Customers | Revenue | Revenue per Customer |
|---|---|---|
| 3259 | 563 K | 172.75 |

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## Customer Analysis — Business Question

**Question - 8** Are we growing because we are acquiring more customers, or because existing customers are buying more?

### SQL Query

1

```sql
with customer_first_order As (
    select
        c.customer_unique_id,
        min(o.purchase_date) As First_order_date
    from olist_dwh.gold.dim_orders o
    left join olist_dwh.gold.dim_customers c
        on o.customer_sk = c.customer_sk
    group by c.customer_unique_id
),
customer_year As (
    select
        year(o.purchase_date) as year,
        c.customer_unique_id,
        cfo.First_order_date,
        count(distinct o.order_sk) as Total_Orders,
        Sum(oi.revenue) As Total_Revenue
    from olist_dwh.gold.dim_orders o
    left join olist_dwh.gold.fact_order_items oi
        on o.order_sk = oi.order_sk
    left join olist_dwh.gold.dim_customers c
        on o.customer_sk = c.customer_sk
    left join customer_first_order cfo
        on c.customer_unique_id = cfo.customer_unique_id
    group by
        year(o.purchase_date),
        c.customer_unique_id,
        cfo.First_order_date
),
Customer_type as (
    select
        year,
        customer_unique_id,
        Total_orders,
        Total_Revenue,
        case
           when year(First_order_date) = year then 'New Customer'
           else 'Repeat Customer'
        end as Customer_type
    from customer_year
)
```

2

```sql
select
    year,
    count(distinct case
        when Customer_type = 'New Customer' then customer_unique_id
        End) As New_customers,
    Sum( case
        when Customer_type = 'New Customer' then Total_Revenue
        else 0
        End) As New_Customers_Revenue,
    Sum( case
        when Customer_type = 'New Customer' then Total_orders
        else 0
        End) As New_Customers_Orders,


    count(distinct case
        when Customer_type = 'Repeat Customer' then customer_unique_id
        End) As Repeat_customers,
    Sum( case
        when Customer_type = 'Repeat Customer' then Total_Revenue
        else 0
        End) As Repeat_Customers_Revenue,
    Sum( case
        when Customer_type = 'Repeat Customer' then Total_orders
        else 0
        End) As Repeat_Customers_Orders

from Customer_type
where year <> 2016
group by year
order by year;
```

### In 2017

- New Customers: 44333
- New Customers Revenue: 7.16 M
- New Customers Orders: 45091
- Repeat Customers: 8
- Repeat Customers Revenue: 1.63 K
- Repeat Customers Orders: 10

### In 2018

- New Customers: 51434
- New Customers Revenue: 8.35 M
- New Customers Orders: 52309
- Repeat Customers: 1536
- Repeat Customers Revenue: 271.05 K
- Repeat Customers Orders: 1702

## Customer Analysis — Business Question

### Question - 9 Which customer segments generate the most revenue?

### SQL Query

1
```sql
with customers_stutas As (
    select
        c.customer_unique_id,
        MAX(o.purchase_date) As Last_order_date,
        (select MAX(purchase_date) from olist_dwh.gold.dim_orders) AS Last_Date,
        DATEDIFF(
            DAY,
            MAX(o.purchase_date),
            (select MAX(purchase_date) from olist_dwh.gold.dim_orders)
            ) As Recency,
        COUNT(Distinct o.order_sk) As Frequency,
        SUM(op.payment_value) As Monetary
    from olist_dwh.gold.dim_orders o
    left join olist_dwh.gold.fact_order_payments op
        on o.order_sk = op.order_sk
    left join olist_dwh.gold.dim_customers c
        on o.customer_sk = c.customer_sk
    group by c.customer_unique_id
),
```

2
```sql
RFM_score As(
    select
        customer_unique_id,
        Monetary,
        CASE
            WHEN Recency <= 60 THEN 5
            WHEN Recency <= 90 THEN 4
            WHEN Recency <= 150 THEN 3
            WHEN Recency <= 365 THEN 2
            ELSE 1
        END As R_score,

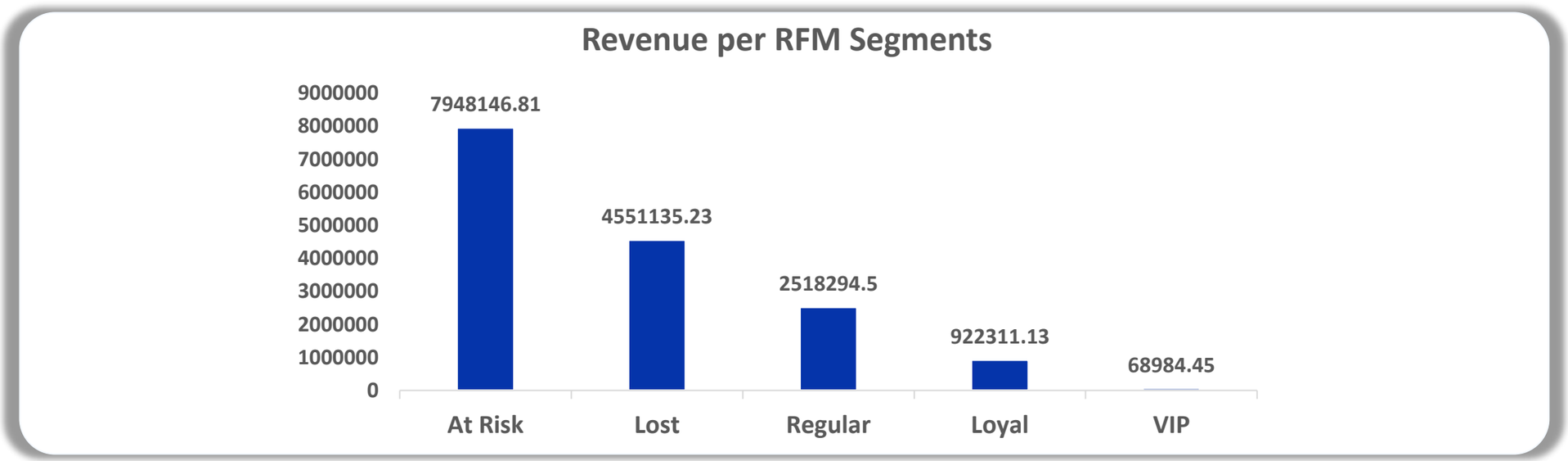
        CASE
            WHEN Frequency <= 1 THEN 1
            WHEN Frequency = 2 THEN 2
            WHEN Frequency = 3 THEN 3
            WHEN Frequency = 4 THEN 4
            ELSE 5
        END As F_score,

        CASE
            WHEN Monetary <= 100 THEN 1
            WHEN Monetary <= 200 THEN 2
            WHEN Monetary <= 500 THEN 3
            WHEN Monetary <= 1000 THEN 4
            ELSE 5
        END As M_score

    from customers_stutas
),
```

3
```sql
customers_segment As (
select
    customer_unique_id,
    Monetary as Total_Revenue,
    CASE
        WHEN R_score = 5  AND( F_score >= 3 OR M_score >= 4 ) THEN 'VIP'
        WHEN R_score >= 4 AND( F_score >= 2 OR M_score >= 3 ) THEN 'Loyal'
        WHEN R_score >= 3 THEN 'Regular'
        WHEN R_score = 2 THEN 'At Risk'
        WHEN R_score = 1 THEN 'Lost'
        ELSE 'Regular'
    END As customer_segment
from RFM_score
)

select
    customer_segment,
    COUNT(*) As customer_count,
    CONCAT( CAST(
        COUNT(*) * 100.00 / SUM(COUNT(*)) over()
        As decimal(10,2)
    ), '%') As customer_percentage,
    SUM(Total_Revenue) As Total_Revenue
from customers_segment
group by customer_segment
order by customer_count DESC;
```



# Business Questions

TOPIC 02 OF 08

## 2. Sellers Performance

Revenue by Seller, Seller Performance, Review Performance, Order Volume, Late Delivery

**Proceed to Analytical Questions & Queries** ➡

## Sellers Performance — Business Question

**Question - 1** Which sellers have the highest order volume?

### SQL Query

```sql
select
    Top 10
    seller_sk,
    COUNT(DISTINCT order_sk) As Total_Orders,
    SUM(revenue) As Total_Revenue,
    CAST(
        SUM(revenue)
        / COUNT(DISTINCT order_sk)
        As decimal(10,2)
    ) AOV_per_Seller
from olist_dwh.gold.fact_order_items
group by seller_sk
having
    COUNT(DISTINCT order_sk) > 100
    AND SUM(revenue) > 10000
order by AOV_per_Seller DESC;
```

| Seller SK | Total Orders | Total Revenue | AOV per Seller |
|---|---|---|---|
| 902 | 358 | 235856.68 | 658.82 |
| 69 | 336 | 182754.05 | 543.91 |
| 2242 | 166 | 83189.65 | 501.14 |
| 2387 | 117 | 57440.94 | 490.95 |
| 10 | 187 | 79261.6 | 423.86 |
| 1648 | 125 | 49531.57 | 396.25 |
| 1264 | 158 | 61679.33 | 390.38 |
| 2341 | 325 | 125816.92 | 387.13 |
| 1993 | 174 | 61829.81 | 355.34 |
| 558 | 585 | 204084.73 | 348.86 |

## Sellers Performance — Business Question

**Question - 2** Which sellers have the best customer ratings ?

### SQL Query

```sql
WITH Seller_Orders As (
    select DISTINCT
        seller_sk,
        order_sk
    from olist_dwh.gold.fact_order_items
)

select
    Top 5
    so.seller_sk,
    COUNT(DISTINCT so.order_sk) As Total_Orders,
    CAST(
        AVG(orv.review_score)
          AS decimal(10,2)
    ) As Avg_review_score
from Seller_Orders so
join olist_dwh.gold.dim_order_reviews orv
    on so.order_sk = orv.order_sk
group by so.seller_sk
having COUNT(DISTINCT so.order_sk) >= 100
order by Avg_review_score DESC;
```

| | Seller SK | Orders | Avg Review Score |
|---|---|---|---|
| 1 | 550 | 110 | 4.0 |
| 2 | 865 | 106 | 4.0 |
| 3 | 716 | 208 | 4.0 |
| 4 | 1303 | 245 | 4.0 |
| 5 | 284 | 317 | 4.0 |

## Sellers Performance — Business Question

**Question - 3** Are high revenue sellers also high performing operationally ?

### SQL Query

1

```sql
With Seller_Orders As (
    select
        oi.seller_sk,
        o.order_sk,
        SUM(oi.revenue) As Order_Revenue,
        CASE
            WHEN o.delivered_customer_date > o.expected_delivery_date THEN 'Delayed'
            WHEN o.delivered_customer_date < o.expected_delivery_date THEN 'Early'
            ELSE 'On Time'
        END As Order_Status
    from olist_dwh.gold.fact_order_items oi
    left join olist_dwh.gold.dim_orders o
        on oi.order_sk =o.order_sk
    where
        o.order_status = 'DELIVERED'
        AND o.delivered_customer_date IS NOT NULL
        AND o.expected_delivery_date IS NOT NULL
    group by
        oi.seller_sk,
        o.order_sk,
        o.delivered_customer_date,
        o.expected_delivery_date
)
```

2

```sql
select
    Top 5
    seller_sk,
    COUNT(order_sk) As Total_Orders,
    SUM(Order_Revenue) As Total_Revenue,
    COUNT(
        CASE
            WHEN Order_Status = 'Delayed' THEN order_sk
        END
    ) As Delayed_Orders,
    SUM(
        CASE
            WHEN Order_Status = 'Delayed' THEN Order_Revenue
            ELSE 0
        END
    ) As Delayed_Revenue,
    CAST(
    COUNT(CASE
            WHEN Order_Status = 'Delayed' THEN order_sk
        END) * 100.00
        / COUNT(order_sk)
          AS decimal(10,2)
    ) As Delayed_Delivery_Rate,
    CAST(
    SUM(CASE
            WHEN Order_Status = 'Delayed' THEN Order_Revenue
            ELSE 0
        END) * 100.00
        / sUM(Order_Revenue)
          AS decimal(10,2)
    ) As Delayed_Revenue_Rate
from Seller_Orders so
group by seller_sk
having COUNT(order_sk) >= 100
order by Total_Revenue DESC;
```

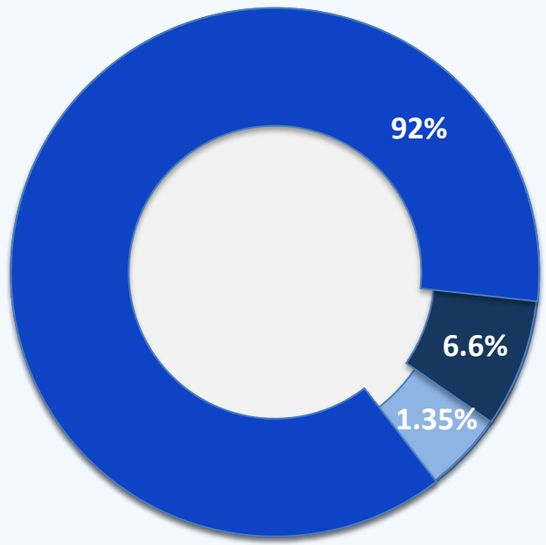
| Seller SK | Total Orders | Total Revenue | Delayed Orders | Delayed Revenue | Delayed Delivery Rate | Delayed Revenue Rate |
|---|---|---|---|---|---|---|
| 2618 | 1094 | 240748.64 | 64 | 13014.07 | 5.85 | 5.41 |
| 902 | 349 | 232293.91 | 32 | 21974.58 | 9.17 | 9.46 |
| 1183 | 952 | 229535.5 | 73 | 17522.26 | 7.67 | 7.63 |
| 2464 | 1746 | 227706.55 | 128 | 17915.91 | 7.33 | 7.87 |
| 558 | 568 | 199446.77 | 42 | 13988.66 | 7.39 | 7.01 |

## Sellers Performance — Business Question

**Question - 4** Which delivery performance have most sellers ?

### SQL Query

```sql
With Seller_Orders As (
    select Distinct
        oi.seller_sk,
        o.order_sk,
        CASE
            WHEN o.delivered_customer_date > o.expected_delivery_date
                THEN 'Delayed'
            WHEN o.delivered_customer_date < o.expected_delivery_date
                THEN 'Early'
            ELSE 'On Time'
        END As Delivery_Performance,
        orv.review_score
    from olist_dwh.gold.fact_order_items oi
    left join olist_dwh.gold.dim_orders o
        on oi.order_sk =o.order_sk
    join olist_dwh.gold.dim_order_reviews orv
        on o.order_sk = orv.order_sk
    where
        o.order_status = 'DELIVERED'
        AND o.delivered_customer_date IS NOT NULL
        AND o.expected_delivery_date IS NOT NULL
)
select
    Delivery_Performance,
    COUNT(seller_sk) As Total_Sellers,
    CAST(
        AVG(review_score * 1.0)
          As decimal(10,2)
    ) As Avg_review_score
from Seller_Orders
group by Delivery_Performance
order by Total_Sellers;
```




- **Early Delivery:** **88,663** Sellers was delivered them orders early, with **4.29** avg review.
- **Late Delivery:** **6,416** out of **96,378** Sellers generate Late orders, with **2.27** avg review.
- **On Time:** **1,299** Sellers delivered on time with avg review by **4.04**.

## Sellers Performance — Business Question

**Question - 5** Which sellers have high sales but poor review ?

### SQL Query

```sql
WITH Sellers_Orders As (
    select
        oi.seller_sk,
        o.order_sk,
        SUM(oi.revenue) As Order_Revenue,
        orv.review_score

    from olist_dwh.gold.fact_order_items oi
    left join olist_dwh.gold.dim_orders o
        on oi.order_sk =o.order_sk
    join olist_dwh.gold.dim_order_reviews orv
        on o.order_sk = orv.order_sk
    where
        o.order_status = 'DELIVERED'
        AND o.delivered_customer_date IS NOT NULL
        AND o.expected_delivery_date IS NOT NULL
    group by
        oi.seller_sk,
        o.order_sk,
        orv.review_score
)
Select
    Top 5
    seller_sk,
    COUNT(order_sk) As Total_Orders,
    SUM(Order_Revenue) As Total_Revenue,
    CAST(
        AVG(review_score)
          As decimal(10,2)
    ) As Avg_review_score
from Sellers_Orders
group by seller_sk
having
    COUNT(order_sk) >= 100
    AND CAST(AVG(review_score)
    As decimal(10,2)) < 4
order by
    Total_Revenue DESC,
    Avg_review_score ASC;
```

| | Seller SK | Orders | Revenue | Avg Review Score |
|---|---|---|---|---|
| 1 | 902 | 110 | 229,624.74 | 4.0 |
| 2 | 424 | 106 | 56,767.21 | 4.0 |
| 3 | 1882 | 208 | 26,314.19 | 4.0 |
| 4 | 1042 | 245 | 23,388.07 | 4.0 |
| 5 | 1099 | 317 | 20,424.73 | 4.0 |

## Sellers Performance — Business Question

**Question - 6** Which sellers should receive operational attention ?

### SQL Query

1

```sql
WITH Sellers_Orders As (
    select
        oi.seller_sk,
        o.order_sk,
        SUM(oi.revenue) As Order_Revenue,
        CASE
            WHEN o.delivered_customer_date > o.expected_delivery_date THEN 'Delayed'
            WHEN o.delivered_customer_date < o.expected_delivery_date THEN 'Early'
            ELSE 'On Time'
        END As Order_Status,
        orv.review_score
    from olist_dwh.gold.fact_order_items oi
    left join olist_dwh.gold.dim_orders o
        on oi.order_sk =o.order_sk
    join olist_dwh.gold.dim_order_reviews orv
        on o.order_sk = orv.order_sk
    where
        o.order_status = 'DELIVERED'
        AND o.delivered_customer_date IS NOT NULL
        AND o.expected_delivery_date IS NOT NULL
    group by
        oi.seller_sk,
        o.order_sk,
        orv.review_score,
        o.delivered_customer_date,
        o.expected_delivery_date
),
```

2

```sql
Seller_Orders_Status As (
    Select
        seller_sk,
        COUNT(order_sk) As Total_Orders,
        SUM(Order_Revenue) As Total_Revenue,
        COUNT(
            CASE
                WHEN Order_Status = 'Delayed' THEN order_sk
            END
        ) As Delayed_Orders,
        SUM(
            CASE
                WHEN Order_Status = 'Delayed' THEN Order_Revenue
                ELSE 0
            END
        ) As Delayed_Revenue,
        CAST(
            COUNT(CASE
                WHEN Order_Status = 'Delayed' THEN order_sk
            END) * 100.00
            / COUNT(order_sk)
              AS decimal(10,2)
        ) As Delayed_Delivery_Rate,
        CAST(
            SUM(CASE
                WHEN Order_Status = 'Delayed' THEN Order_Revenue
                ELSE 0
            END) * 100.00
            / SUM(Order_Revenue)
              AS decimal(10,2)
        ) As Delayed_Revenue_Rate,
        CAST(
            AVG(review_score)
              As decimal(10,2)
        ) As Avg_review_score
    from Sellers_Orders
    group by seller_sk
)
select *
from Seller_Orders_Status
where Total_Orders >= 100
      AND Delayed_Delivery_Rate > 10
      AND  Delayed_Revenue_Rate > 10;
```

| Seller SK | Total Orders | Total Revenue | Delayed Orders | Delayed Revenue | Delayed Delivery Rate | Delayed Revenue Rate | Avg review score |
|---|---|---|---|---|---|---|---|
| 277 | 128 | 16075.14 | 15 | 1740.6 | 11.72 | 10.83 | 3.0 |
| 453 | 141 | 15789.68 | 15 | 1755.24 | 10.64 | 11.12 | 3.0 |
| 665 | 113 | 21768.27 | 14 | 2914.5 | 12.39 | 13.39 | 4.0 |
| 2622 | 222 | 9132.8 | 24 | 1057.88 | 10.81 | 11.58 | 4.0 |
| 2497 | 172 | 14193.59 | 22 | 2058.48 | 12.79 | 14.5 | 3.0 |

## Sellers Performance — Business Question

### Question - 7 Are our top sellers actually the best performance sellers ?

1

```sql
WITH Sellers_Orders As (
    select
        oi.seller_sk,
        o.order_sk,
        o.order_status,
        COALESCE(SUM(oi.revenue),0) As Order_Value,
        CASE
            WHEN o.order_status = 'DELIVERED'
                 AND o.delivered_customer_date > o.expected_delivery_date
                THEN 'Delayed'
            WHEN o.order_status = 'DELIVERED'
                 AND o.delivered_customer_date < o.expected_delivery_date
                THEN 'Early'
            WHEN o.order_status = 'DELIVERED'
                THEN 'On Time'
            ELSE 'N/A'
        END As Delivery_Performance,
        orv.review_score
    from olist_dwh.gold.fact_order_items oi
    left join olist_dwh.gold.dim_orders o
        on oi.order_sk =o.order_sk
    join olist_dwh.gold.dim_order_reviews orv
        on o.order_sk = orv.order_sk
    group by
        oi.seller_sk,
        o.order_sk,
        o.order_status,
        o.delivered_customer_date,
        o.expected_delivery_date,
        orv.review_score
)
```

2

```sql
Select
    Top 5
    seller_sk,
    COUNT(order_sk) As Total_Orders,
    SUM(Order_Value) As Total_Revenue,
    ------------------------------------------------
    -- Delivery Volume
    ------------------------------------------------
    COUNT(
        CASE
            WHEN Order_Status = 'DELIVERED' THEN order_sk
        END
    ) As Delivered_Orders,
    COUNT(
        CASE
            WHEN Delivery_Performance = 'Delayed' THEN order_sk
        END
    ) As Delayed_Orders,
    SUM(
        CASE
            WHEN Delivery_Performance = 'Delayed' THEN Order_Value
            ELSE 0
        END
    ) As Delayed_Revenue,
    ------------------------------------------------
    -- Delivery Performance
    ------------------------------------------------
    CAST(
        COUNT(CASE
            WHEN Delivery_Performance = 'Delayed' THEN order_sk
        END) * 100.00
        /
        NULLIF(
            COUNT(
                CASE
                    WHEN Order_Status = 'DELIVERED' THEN order_sk
                END
            ), 0
        )
        AS decimal(10,2)
    ) As Delayed_Delivery_Rate,
    CAST(
        SUM(CASE
            WHEN Delivery_Performance = 'Delayed' THEN Order_Value
            ELSE 0
        END) * 100.00
        /
        NULLIF(
            SUM(
                CASE
                    WHEN Order_Status = 'DELIVERED' THEN Order_Value
                    ELSE 0
                END
            ), 0
        )
        AS decimal(10,2)
    ) As Delayed_Revenue_Rate,
```

3

```sql
    ------------------------------------------------
    -- Customer Satisfaction
    ------------------------------------------------
    CAST(
        AVG(review_score)
          As decimal(10,2)
    ) As Avg_review_score,
    ------------------------------------------------
    -- Other Order Statuses
    ------------------------------------------------
    COUNT(
        CASE
            WHEN Order_Status = 'INVOICED' THEN order_sk
        END
    ) As Invoiced_Orders,
    COUNT(
        CASE
            WHEN Order_Status = 'PROCESSING' THEN order_sk
        END
    ) As Processing_Orders,
    COUNT(
        CASE
            WHEN Order_Status = 'SHIPPED' THEN order_sk
        END
    ) As Shipped_Orders,
    COUNT(
        CASE
            WHEN Order_Status = 'CANCELED' THEN order_sk
        END
    ) As Canceled_Orders
from Sellers_Orders
group by seller_sk
having COUNT(order_sk) > = 100
order by Total_Revenue DESC;
```

| seller SK | Total Orders | Total Revenue | Delivered Orders | Delayed Orders | Delayed Revenue | Delayed Delivery Rate | Delayed Revenue Rate | Avg review score | Shipped Orders | Canceled Orders |
|---|---|---|---|---|---|---|---|---|---|---|
| 2618 | 1129 | 249,135.73 | 1091 | 63 | 12,853.31 | 5.77 | 5.35 | 4.0 | 13 | 3 |
| 1183 | 975 | 237,052.58 | 945 | 72 | 17,422.37 | 7.62 | 7.67 | 4.0 | 11 | 10 |
| 2464 | 1795 | 233,489.06 | 1737 | 126 | 17,711.01 | 7.25 | 7.84 | 4.0 | 21 | 7 |
| 902 | 354 | 232,987.72 | 346 | 32 | 21,974.58 | 9.25 | 9.57 | 3.0 | 1 | 1 |
| 558 | 583 | 203,627.55 | 567 | 42 | 13,988.66 | 7.41 | 7.02 | 4.0 | 5 | 3 |

## Business Questions



TOPIC 03 OF 08

# 3. Products & Catalog

Top Products, Product Performance, Category Revenue, Products Review

**Proceed to Analytical Questions & Queries** ➡

## Products & Catalog — Business Question

**Question - 1** Which product categories drive revenue ?

### SQL Query

```
select
    TOP 10
    COALESCE(pc.product_category_name_engilsh, 'Others') As Product_Category,
    SUM(oi.revenue) As Total_Revenue
from olist_dwh.gold.fact_order_items oi
left join  olist_dwh.gold.dim_products p
    on oi.product_sk = p.product_sk
left join  olist_dwh.gold.dim_product_categories pc
    on p.product_category_id = pc.product_category_id
group by COALESCE(pc.product_category_name_engilsh, 'Others')
order by Total_Revenue desc;
```

| Product Category | Total Revenue |
|---|---|
| Health Beauty | 1,441,248.07 |
| Watches Gifts | 1,305,541.61 |
| Bed Bath Table | 1,241,681.72 |
| Sports Leisure | 1,156,656.48 |
| Computers Accessories | 1,059,272.40 |
| Furniture Decor | 902,511.79 |
| Housewares | 778,397.77 |
| Cool Stuff | 719,329.95 |
| Auto | 685,384.32 |
| Garden Tools | 584,219.21 |

## Products & Catalog — Business Question

**Question - 2** Which categories generate the highest number of orders ?

### SQL Query

```sql
select
    Top 10
    pc.product_category_name_engilsh As product_category,
    COUNT(Distinct o.order_sk) As Total_Orders
from olist_dwh.gold.fact_order_items oi
left join olist_dwh.gold.dim_orders o
    on oi.order_sk = o.order_sk
left join olist_dwh.gold.dim_products p
    on oi.product_sk = p.product_sk
left join olist_dwh.gold.dim_product_categories pc
    on p.product_category_id = pc.product_category_id
group by pc.product_category_name_engilsh
order by Total_Orders DESC;
```

| Product Category | Total Orders |
|---|---|
| Bed Bath Table | 9,417 |
| Health Beauty | 8,836 |
| Sports Leisure | 7,720 |
| Computers Accessories | 6,689 |
| Furniture Decor | 6,449 |
| Housewares | 5,884 |
| Watches Gifts | 5,624 |
| Telephony | 4,199 |
| Auto | 3,897 |
| Toys | 3,886 |

## Products & Catalog — Business Question

**Question - 3** Which categories have the highest average order value ?

### SQL Query

```sql
select
    Top 10
    pc.product_category_name_engilsh As product_category,
    COUNT(Distinct o.order_sk) As Total_Orders,
    SUM(oi.revenue) As Total_Revenue,
    CAST(SUM(oi.revenue) / COUNT(Distinct o.order_sk) As decimal(10,2)) As AOV
from olist_dwh.gold.fact_order_items oi
left join olist_dwh.gold.dim_orders o
    on oi.order_sk = o.order_sk
left join olist_dwh.gold.dim_products p
    on oi.product_sk = p.product_sk
left join olist_dwh.gold.dim_product_categories pc
    on p.product_category_id = pc.product_category_id
group by pc.product_category_name_engilsh
order by AOV DESC;
```

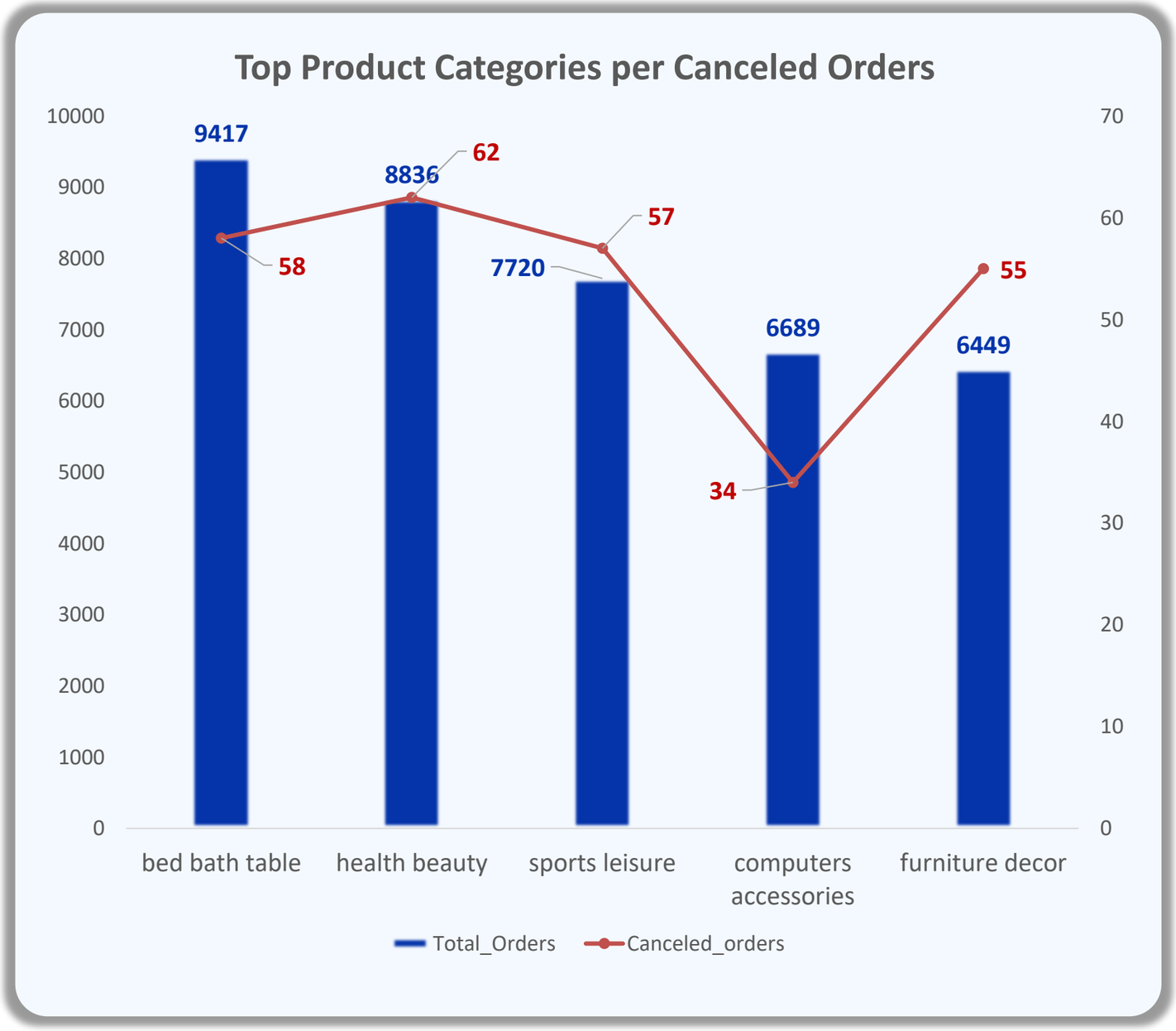
| Product Category | Total Orders | Total Revenue | AOV |
|---|---|---|---|
| Computers | 181 | 232,799.43 | 1,286.18 |
| Small Appliances Home Oven And Coffee | 75 | 50,193.57 | 669.25 |
| Home Appliances 2 | 234 | 123,917.92 | 529.56 |
| Agro Industry And Commerce | 182 | 78,374.07 | 430.63 |
| Musical Instruments | 628 | 210,137.37 | 334.61 |
| Small Appliances | 630 | 206,668.83 | 328.05 |
| Fixed Telephony | 217 | 64,220.81 | 295.95 |
| Office Furniture | 1,273 | 342,532.65 | 269.08 |
| Construction Tools Safety | 167 | 44,463.62 | 266.25 |
| Furniture Bedroom | 95 | 24,661.01 | 259.59 |

## Products & Catalog — Business Question

**Question - 4** Which products have unusually high cancellation behavior ?

### SQL Query

```sql
select
    Top 5
    pc.product_category_name_engilsh As product_category,
    COUNT(Distinct o.order_sk) As Total_Orders,
    SUM(
        CASE
            WHEN o.order_status = 'CANCELED' THEN 1
            ELSE 0
    END) As Canceled_orders
from olist_dwh.gold.fact_order_items oi
left join olist_dwh.gold.dim_orders o
    on oi.order_sk = o.order_sk
left join olist_dwh.gold.dim_products p
    on oi.product_sk = p.product_sk
left join olist_dwh.gold.dim_product_categories pc
    on p.product_category_id = pc.product_category_id
group by pc.product_category_name_engilsh
order by Total_Orders DESC;
```



## Products & Catalog — Business Question

**Question - 5** Which products contribute the most revenue growth ?

**SQL Query**

1

```sql
with product_QUARTERly_revenue As (
    select
        DATETRUNC(QUARTER,o.purchase_date) As QUARTER,
        p.product_sk,
        COALESCE(pc.product_category_name_engilsh, 'Others')  As product_category,
        SUM(oi.revenue) As Total_Revenue
    from olist_dwh.gold.fact_order_items oi
    left join  olist_dwh.gold.dim_products p
        on oi.product_sk = p.product_sk
    left join olist_dwh.gold.dim_product_categories pc
        on p.product_category_id = pc.product_category_id
    left join olist_dwh.gold.dim_orders o
        on oi.order_sk = o.order_sk
    group by
        DATETRUNC(QUARTER,o.purchase_date),
        p.product_sk,
        pc.product_category_name_engilsh
),
```

2

```sql
product_growth As(
    select
        QUARTER,
        product_sk,
        product_category,
        Total_Revenue,
        Lag(Total_Revenue) over(
           partition by product_sk
           order by QUARTER
        ) As Prev_QUARTER_Revenue
    from product_QUARTERly_revenue
)
select
    TOP 5
    QUARTER,
    product_sk,
    product_category,
    Total_Revenue As Current_QUARTER_Revenue,
    Prev_QUARTER_Revenue,
    Total_Revenue - Prev_QUARTER_Revenue As Revenue_Growth,
    CAST(
    (Total_Revenue - Prev_QUARTER_Revenue)
    / NULLIF(Prev_QUARTER_Revenue, 0) * 100
    As decimal(10,2)
    ) As Revenue_Growth_percentage
from product_growth
where YEAR(QUARTER) <> 2016 And Prev_QUARTER_Revenue IS NOT NULL
order by Revenue_Growth DESC;
```

| Quarter | Product Sk | Category | Current Quarter Revenue | Prev Quarter Revenue | Revenue Growth | Revenue Growth % |
|---|---|---|---|---|---|---|
| 2018-04-01 | 20013 | Computers | 16,088.72 | 7,593.43 | 8,495.29 | 111.88 |
| 2017-10-01 | 20013 | Computers | 13,329.33 | 5,171.83 | 8,157.50 | 157.73 |
| 2017-10-01 | 19248 | Computers Accessories | 12,359.02 | 4,303.53 | 8,055.49 | 187.18 |
| 2018-01-01 | 17331 | Auto | 8,256.51 | 600.43 | 7,656.08 | 1275.1 |
| 2017-10-01 | 14999 | Watches Gifts | 9,476.17 | 2,038.39 | 7,437.78 | 364.89 |

## Products & Catalog — Business Question

**Question - 6** Which products generate high sales volume but low revenue ?

### SQL Query

1

```sql
with products_performance As (
    select
        p.product_sk,
        COALESCE(pc.product_category_name_engilsh, 'Others')  As product_category,
        COUNT(oi.order_item_sk) As Total_Qty,
        SUM(oi.revenue) As Total_Revenue
    from olist_dwh.gold.fact_order_items oi
    left join  olist_dwh.gold.dim_products p
        on oi.product_sk = p.product_sk
    left join olist_dwh.gold.dim_product_categories pc
        on p.product_category_id = pc.product_category_id
    group by
        p.product_sk,
        pc.product_category_name_engilsh
),
```

2

```sql
product_rank As (
    select
        *,
        CAST(
        PERCENT_RANK( ) over(
        order by Total_Qty
        ) As decimal(5,2))As Qty_percentage,
        CAST(
        PERCENT_RANK( ) over(
        order by Total_Revenue
        ) As decimal(5,2)) As Revenue_percentage
    from products_performance
)
select
    TOP 5
    product_sk,
    product_category,
    Total_Qty,
    Total_Revenue,
    Qty_percentage,
    Revenue_percentage
from product_rank
where Qty_percentage > 0.75
  AND Revenue_percentage < 0.25
order by Total_Qty DESC;
```

| Product Sk | Category | Total Qty | Total Revenue | Qty percentage | Revenue percentage |
|---|---|---|---|---|---|
| 22114 | Housewares | 6 | 71.1 | 0.88 | 0.21 |
| 12515 | Signaling And Security | 6 | 77.34 | 0.88 | 0.23 |
| 26336 | Housewares | 5 | 67.7 | 0.85 | 0.19 |
| 3058 | Sports Leisure | 4 | 73.12 | 0.81 | 0.21 |
| 18005 | Garden Tools | 4 | 80.92 | 0.81 | 0.24 |

## Question - 7 Which categories are growing over time ?

### SQL Query

1

```sql
with product_category_QUARTERly_revenue As (
    select
        DATETRUNC(QUARTER,o.purchase_date) As QUARTER,
        COALESCE(pc.product_category_name_engilsh, 'Others')  As product_category,
        SUM(oi.revenue) As Total_Revenue
    from olist_dwh.gold.fact_order_items oi
    left join  olist_dwh.gold.dim_products p
        on oi.product_sk = p.product_sk
    left join olist_dwh.gold.dim_product_categories pc
        on p.product_category_id = pc.product_category_id
    left join olist_dwh.gold.dim_orders o
        on oi.order_sk = o.order_sk
    group by
        DATETRUNC(QUARTER,o.purchase_date),
        pc.product_category_name_engilsh
),
product_category_growth As(
    select
        QUARTER,
        product_category,
        Total_Revenue,
        Lag(Total_Revenue) over(
           partition by product_category
           order by QUARTER
        ) As Prev_QUARTER_Revenue
    from product_category_QUARTERly_revenue
),
```

2

```sql
product_category_YOY As (
    select
        QUARTER,
        product_category,
        Total_Revenue As Current_QUARTER_Revenue,
        Prev_QUARTER_Revenue,
        Total_Revenue - Prev_QUARTER_Revenue As Revenue_Growth,
        CAST(
        (Total_Revenue - Prev_QUARTER_Revenue)
        / NULLIF(Prev_QUARTER_Revenue, 0) * 100
        As decimal(10,2)
        ) As Revenue_Growth_percentage
    from product_category_growth
    where YEAR(QUARTER) <> 2016 AND Prev_QUARTER_Revenue IS NOT NULL
),
final_performance As (
select
    *,
    CASE
        WHEN Revenue_Growth > 0 THEN 'Growing'
        WHEN Revenue_Growth < 0 THEN 'Declining'
        ELSE 'Stable'
    END AS Trend
from product_category_YOY
)
select TOP 5 *
from final_performance
where Trend = 'Growing'   -- Declining / Stable
order by Revenue_Growth DESC;
```

| Quarter | Category | Cur Quarter Revenue | Prev Quarter Revenue | Revenue Growth | Revenue Growth % | Trend |
|---|---|---|---|---|---|---|
| 2017-10-01 | Watches Gifts | 251,183.05 | 150,222.43 | 100,960.62 | 67.21 % | Growing |
| 2017-10-01 | Computers Accessories | 210,857.27 | 117,153.53 | 93,703.74 | 79.98 % | Growing |
| 2017-04-01 | Health Beauty | 142,290.76 | 66,705.69 | 75,585.07 | 113.31 % | Growing |
| 2018-01-01 | Health Beauty | 317,084.22 | 244,618.60 | 72,465.62 | 29.62 % | Growing |
| 2017-01-01 | Watches Gifts | 73,611.04 | 5,585.06 | 68,025.98 | 1218 % | Growing |

## Products & Catalog — Business Question

### Question - 8 Which categories are declining over time ?

### SQL Query

1

```sql
with product_category_QUARTERly_revenue As (
    select
        DATETRUNC(QUARTER,o.purchase_date) As QUARTER,
        COALESCE(pc.product_category_name_engilsh, 'Others')  As product_category,
        SUM(oi.revenue) As Total_Revenue
    from olist_dwh.gold.fact_order_items oi
    left join  olist_dwh.gold.dim_products p
        on oi.product_sk = p.product_sk
    left join olist_dwh.gold.dim_product_categories pc
        on p.product_category_id = pc.product_category_id
    left join olist_dwh.gold.dim_orders o
        on oi.order_sk = o.order_sk
    group by
        DATETRUNC(QUARTER,o.purchase_date),
        pc.product_category_name_engilsh
),
product_category_growth As(
    select
        QUARTER,
        product_category,
        Total_Revenue,
        Lag(Total_Revenue) over(
           partition by product_category
           order by QUARTER
        ) As Prev_QUARTER_Revenue
    from product_category_QUARTERly_revenue
),
```

2

```sql
product_category_YOY As (
    select
        QUARTER,
        product_category,
        Total_Revenue As Current_QUARTER_Revenue,
        Prev_QUARTER_Revenue,
        Total_Revenue - Prev_QUARTER_Revenue As Revenue_Growth,
        CAST(
        (Total_Revenue - Prev_QUARTER_Revenue)
        / NULLIF(Prev_QUARTER_Revenue, 0) * 100
        As decimal(10,2)
        ) As Revenue_Growth_percentage
    from product_category_growth
    where YEAR(QUARTER) <> 2016 AND Prev_QUARTER_Revenue IS NOT NULL
),
final_performance As (
select
    *,
    CASE
        WHEN Revenue_Growth > 0 THEN 'Growing'
        WHEN Revenue_Growth < 0 THEN 'Declining'
        ELSE 'Stable'
    END AS Trend
from product_category_YOY
)
select TOP 5 *
from final_performance
where Trend = 'Declining'
order by
    Revenue_Growth;
```

| Quarter | Category | Cur Quarter Revenue | Prev Quarter Revenue | Revenue Growth | Revenue Growth % | Trend |
|---|---|---|---|---|---|---|
| 2018-10-01 | Watches Gifts | 144.56 | 169986.06 | -169841.5 | -99.91 | Declining |
| 2018-10-01 | Bed Bath Table | 76.15 | 166801.41 | -166725.26 | -99.95 | Declining |
| 2018-07-01 | Health Beauty | 184166.27 | 293507.05 | -109340.78 | -37.25 | Declining |
| 2018-07-01 | Sports Leisure | 142037.58 | 245490.62 | -103453.04 | -42.14 | Declining |
| 2018-07-01 | Computers Accessories | 121608.67 | 223098.53 | -101489.86 | -45.49 | Declining |

## Products & Catalog — Business Question

**Question - 9** Are there products with high demand but poor customer rating ?

### SQL Query

```sql
select
    p.product_sk,
    COALESCE(pc.product_category_name_engilsh, 'Others')  As product_category,
    COUNT(distinct oi.order_sk) As Total_Orders,
    SUM(oi.revenue) As Total_Revenue,
    CAST(
    AVG(orv.review_score)
    As decimal(10,2)
    ) As AVG_Review_Score
from olist_dwh.gold.fact_order_items oi
left join  olist_dwh.gold.dim_products p
    on oi.product_sk = p.product_sk
left join olist_dwh.gold.dim_product_categories pc
    on p.product_category_id = pc.product_category_id
left join olist_dwh.gold.dim_order_reviews orv
    on oi.order_sk = orv.order_sk
group by
    p.product_sk,
    COALESCE(pc.product_category_name_engilsh, 'Others')
having
    COUNT(distinct oi.order_sk) > 100
    AND AVG(orv.review_score) < 4.0
order by Total_Orders desc;
```

| product SK | Category | Total Orders | Total Revenue | Avg Review Score |
|---|---|---|---|---|
| 20412 | Garden Tools | 311 | 28,682.68 | 3.00 |
| 19687 | Others | 194 | 27,286.05 | 3.00 |
| 22258 | Housewares | 160 | 14,258.97 | 3.00 |
| 9171 | Health Beauty | 151 | 59,093.99 | 3.00 |
| 17803 | Fashion Bags Accessories | 138 | 8,618.47 | 3.00 |
| 26013 | Perfumery | 134 | 9,525.10 | 3.00 |
| 32285 | Health Beauty | 131 | 5,484.97 | 3.00 |
| 20826 | Garden Tools | 129 | 20,009.36 | 3.00 |
| 30668 | Health Beauty | 120 | 12,514.04 | 3.00 |
| 14909 | Garden Tools | 114 | 25,886.26 | 3.00 |
| 12088 | Electronics | 104 | 3,109.10 | 3.00 |

## Products & Catalog — Business Question

**Question - 10** Which products should olist prioritize based on revenue, demand, customer satisfaction ?

### SQL Query

```sql
select
    TOP 10
    p.product_sk,
    COALESCE(pc.product_category_name_engilsh, 'Others')  As product_category,
    COUNT(distinct oi.order_sk) As Total_Orders,
    SUM(oi.revenue) As Total_Revenue,
    CAST(
    AVG(orv.review_score)
    As decimal(10,2)
    ) As AVG_Review_Score
from olist_dwh.gold.fact_order_items oi
left join  olist_dwh.gold.dim_products p
    on oi.product_sk = p.product_sk
left join olist_dwh.gold.dim_product_categories pc
    on p.product_category_id = pc.product_category_id
left join olist_dwh.gold.dim_order_reviews orv
    on oi.order_sk = orv.order_sk
group by
    p.product_sk,
    COALESCE(pc.product_category_name_engilsh, 'Others')
having
    COUNT(distinct oi.order_sk) >= 100
    AND SUM(oi.revenue) >= 10000
    AND AVG(orv.review_score) >= 4.0
order by Total_Orders desc;
```

| product SK | Category | Total Orders | Total Revenue | Avg Review Score |
|---|---|---|---|---|
| 32731 | Bed Bath Table | 467 | 51,071.60 | 4.00 |
| 3549 | Furniture Decor | 431 | 44,820.76 | 4.00 |
| 4170 | Garden Tools | 352 | 34,201.26 | 4.00 |
| 31360 | Computers Accessories | 323 | 60,976.03 | 4.00 |
| 23863 | Watches Gifts | 306 | 39,957.93 | 4.00 |
| 27668 | Garden Tools | 291 | 27,984.40 | 4.00 |
| 22217 | Garden Tools | 287 | 27,268.23 | 4.00 |
| 24819 | Health Beauty | 269 | 10,063.11 | 4.00 |
| 28892 | Health Beauty | 259 | 26,659.74 | 4.00 |
| 19248 | Computers Accessories | 255 | 48,212.22 | 4.00 |

## Business Questions



TOPIC 04 OF 08

# 4. Orders & Logistics

Order Trends, Late Deliveries, Items per Order, Delivery Time, Cancellation Rate

**Proceed to Analytical Questions & Queries** ➡

## Orders & Logistics — Business Question

**Question - 1** What percentage of orders are delivered late ?

### SQL Query

```sql
select
    COUNT(*) As Delivered_Orders,
    SUM(
        CASE
            WHEN delivered_customer_date > expected_delivery_date THEN 1
            ELSE 0
        END
    ) As Late_Orders,

    CONCAT(
        CAST(
            SUM(
                CASE
                    WHEN delivered_customer_date > expected_delivery_date THEN 1
                    ELSE 0
                END
            ) * 100.00
            / COUNT(*)
            As decimal(10,2)
        )
        , '%'
    ) As Late_Orders_Pct

from olist_dwh.gold.dim_orders
where
    order_status = 'DELIVERED'
    AND delivered_customer_date IS NOT NULL
    AND expected_delivery_date IS NOT NULL;
```

| Delivered Orders | Late Orders | Late Orders Pct |
|---|---|---|
| 96,470 | 6,519 | 6.76 % |

## Question - 2 Which states have the longest delivery times ?

### SQL Query

```sql
select
    TOP 10
    COALESCE(l.state, 'N/A') As State,
    COUNT(o.order_sk) As Delivered_Orders,
    CAST(
        AVG(
            DATEDIFF(
            DAY,
            o.purchase_date,
            o.delivered_customer_date
            ) * 1.0
        )
        As decimal(10,2)
    ) As Avg_Delivery_Days

from olist_dwh.gold.dim_orders o
left join olist_dwh.gold.dim_customers c
    on o.customer_sk = c.customer_sk
left join olist_dwh.gold.dim_geo_location l
    on c.geo_location_sk = l.geo_location_sk
where
    o.order_status = 'DELIVERED'
    AND o.purchase_date IS NOT NULL
    AND o.delivered_customer_date IS NOT NULL
group by COALESCE(l.state, 'N/A')
order by Avg_Delivery_Days DESC;
```

| State | Delivered Orders | Avg Delivery Days |
|---|---|---|
| Amapa | 65 | 15.88 |
| Amazonas | 140 | 14.98 |
| Roraima | 46 | 14.41 |
| Tocantins | 273 | 14.23 |
| Paraiba | 516 | 14.02 |
| Unknown | 285 | 13.93 |
| Mato Grosso Do Sul | 690 | 12.98 |
| Pernambuco | 1599 | 12.83 |
| Rio Grande Do Norte | 465 | 12.76 |
| Rio Grande Do Sul | 5309 | 12.75 |

## Question - 3 Which sellers have the highest late delivery rates ?

### SQL Query

```sql
select
    TOP 10
    s.seller_sk,
    COUNT(o.order_sk) As Delivered_Orders,
    CAST(
        AVG(
            DATEDIFF(
            DAY,
            o.purchase_date,
            o.delivered_customer_date
            ) * 1.0
        )
        As decimal(10,2)
    ) As Avg_Delivery_Days
from olist_dwh.gold.dim_orders o
left join olist_dwh.gold.fact_order_items oi
    on o.order_sk = oi.order_sk
left join olist_dwh.gold.dim_sellers s
    on oi.seller_sk = s.seller_sk
where
    o.order_status = 'DELIVERED'
    AND o.purchase_date IS NOT NULL
    AND o.delivered_customer_date IS NOT NULL
group by s.seller_sk
order by Avg_Delivery_Days DESC;
```

| seller sk | Delivered Orders | Avg Delivery Days |
|---|---|---|
| 92 | 1 | 67 |
| 1154 | 1 | 49 |
| 3019 | 1 | 49 |
| 2805 | 1 | 45 |
| 184 | 5 | 44.4 |
| 2240 | 1 | 43 |
| 364 | 4 | 39.5 |
| 947 | 2 | 39 |
| 689 | 2 | 38.5 |
| 1109 | 1 | 38 |

## Question - 4 Is late delivery affecting customer review scores ?

### SQL Query

```sql
WITH Delivery_Reviews As (
    select
        o.order_sk As Delivered_Orders,
        CASE
            WHEN o.delivered_customer_date > o.expected_delivery_date THEN 'Late'
            ELSE 'On Time'
        END As Delivery_Status,
        orv.review_score
    from olist_dwh.gold.dim_orders o
    join olist_dwh.gold.dim_order_reviews orv
        on o.order_sk = orv.order_sk
    where
        o.order_status = 'DELIVERED'
        AND o.expected_delivery_date IS NOT NULL
        AND o.delivered_customer_date IS NOT NULL
)
SELECT
    Delivery_Status,
    review_score,
    COUNT(*) AS Orders,
    SUM(COUNT(*) ) OVER(Partition by Delivery_Status) As Orders_per_Delivery_Status,
    CAST(
        COUNT(*) *100.00
        / SUM(COUNT(*) ) OVER(Partition by Delivery_Status)
        AS decimal(10,2)
    ) As Percentage
FROM Delivery_Reviews
GROUP BY
    Delivery_Status,
    review_score
ORDER BY
    Delivery_Status,
    review_score;
```

| Delivery Status | review score | Orders | Orders per Delivery Status | Percentage |
|---|---|---|---|---|
| Late | 1 | 3430 | 6366 | 53.88 |
| Late | 2 | 549 | 6366 | 8.62 |
| Late | 3 | 694 | 6366 | 10.9 |
| Late | 4 | 646 | 6366 | 10.15 |
| Late | 5 | 1047 | 6366 | 16.45 |
| On Time | 1 | 5921 | 89458 | 6.62 |
| On Time | 2 | 2368 | 89458 | 2.65 |
| On Time | 3 | 7220 | 89458 | 8.07 |
| On Time | 4 | 18249 | 89458 | 20.4 |
| On Time | 5 | 55700 | 89458 | 62.26 |

## Orders & Logistics — Business Question

### Question - 5 Which product categories experience the most delivery delays ?

### SQL Query

1
```sql
WITH Prod_Category_Orders As (
    select
        COALESCE(pc.product_category_id, 0) As product_category_id,
        COALESCE(pc.product_category_name, 'N/A') As Product_Category,
        COUNT(Distinct o.order_sk) As Total_Orders,
        CAST(
            AVG(--
                DATEDIFF(--
                DAY,
                o.purchase_date,
                o.delivered_customer_date
                )
            ) As decimal(10,2)
        ) As Avg_Delivery_Days
    from olist_dwh.gold.dim_orders o
    left join olist_dwh.gold.fact_order_items oi
        on o.order_sk = oi.order_sk
    left join olist_dwh.gold.dim_products p
        on oi.product_sk = p.product_sk
    left join olist_dwh.gold.dim_product_categories pc
        on p.product_category_id = pc.product_category_id
    group by
        COALESCE(pc.product_category_id, 0),
        COALESCE(pc.product_category_name, 'N/A')
),
```

2
```sql
Prod_Category_Deleyed_Orders As (
    select
        COALESCE(pc.product_category_id, 0) As product_category_id,
        COALESCE(pc.product_category_name, 'N/A') As Product_Category,
        COUNT(Distinct o.order_sk) As Delayed_Orders,
        CAST(
            AVG(--
                DATEDIFF(--
                DAY,
                o.expected_delivery_date,
                o.delivered_customer_date
                )
            ) As decimal(10,2)
        ) As Avg_Delay_Days
    from olist_dwh.gold.dim_orders o
    left join olist_dwh.gold.fact_order_items oi
        on o.order_sk = oi.order_sk
    left join olist_dwh.gold.dim_products p
        on oi.product_sk = p.product_sk
    left join olist_dwh.gold.dim_product_categories pc
        on p.product_category_id = pc.product_category_id
    where o.delivered_customer_date > o.expected_delivery_date
    group by
        COALESCE(pc.product_category_id, 0),
        COALESCE(pc.product_category_name, 'N/A')
)
```

3
```sql
select
    TOP 10
    pco.product_category_id AS ID,
    pco.Product_Category,
    pco.Total_Orders,
    pcdo.Delayed_Orders,
    CAST(
        pcdo.Delayed_Orders * 100.00
        / pco.Total_Orders
        As decimal(10,2)
    ) As Delay_Rate,
    pcdo.Avg_Delay_Days
from Prod_Category_Orders pco
left join Prod_Category_Deleyed_Orders pcdo
    on pco.product_category_id = pcdo.product_category_id
where
    pcdo.Delayed_Orders IS NOT NULL
    AND pco.Total_Orders >= 100
order by Delay_Rate desc;
```

| ID | Product Category | Total Orders | Delayed Orders | Delay Rate | Avg Delay Days |
|---|---|---|---|---|---|
| 50 | construcao ferramentas seguranca | 167 | 16 | 9.58 | 12 |
| 49 | livros interesse geral | 512 | 43 | 8.4 | 13 |
| 43 | moveis sala | 422 | 34 | 8.06 | 12 |
| 20 | consoles games | 1062 | 85 | 8 | 11 |
| 11 | alimentos bebidas | 227 | 18 | 7.93 | 22 |
| 22 | fashion calcados | 240 | 19 | 7.92 | 7 |
| 16 | telefonia fixa | 217 | 17 | 7.83 | 13 |
| 55 | artigos de natal | 128 | 10 | 7.81 | 7 |
| 24 | malas acessorios | 1034 | 80 | 7.74 | 10 |
| 54 | pcs | 181 | 14 | 7.73 | 8 |

## Orders & Logistics — Business Question

**Question - 6** How has delivery performance changed over time ?

### SQL Query

```sql
select
    DATETRUNC(
        QUARTER,
        purchase_date
    ) As Quarter,
    COUNT(Distinct order_sk) As Delivered_Orders,
    CAST(
        AVG(--
            DATEDIFF(--
            DAY,
            purchase_date,
            delivered_customer_date
            )
        ) As decimal(10,2)
    ) As Avg_Delivery_Days,
    Count(
        Distinct CASE
            When delivered_customer_date > expected_delivery_date
            Then order_sk
        END ) As Delayed_Orders,
    CAST(
        COUNT(
            DISTINCT CASE
                WHEN delivered_customer_date > expected_delivery_date
                THEN order_sk
            END
        ) * 100.00
        / COUNT(DISTINCT order_sk)
        AS DECIMAL(10,2)
    ) AS Delay_Rate
from olist_dwh.gold.dim_orders
where order_status = 'DELIVERED'
group by DATETRUNC(QUARTER, purchase_date)
order by Quarter;
```

| Quarter | Delivered Orders | Avg Delivery Days | Delayed Orders | Delay Rate |
|---|---|---|---|---|
| 2016-07-01 | 1 | 55 | 1 | 100 % |
| 2016-10-01 | 266 | 19 | 2 | 0.75 % |
| 2017-01-01 | 4949 | 13 | 185 | 3.74 % |
| 2017-04-01 | 8984 | 12 | 351 | 3.91 % |
| 2017-07-01 | 12215 | 11 | 412 | 3.37 % |
| 2017-10-01 | 17280 | 14 | 1494 | 8.65 % |
| 2018-01-01 | 20628 | 15 | 2655 | 12.87 % |
| 2018-04-01 | 19646 | 10 | 820 | 4.17 % |
| 2018-07-01 | 12509 | 8 | 599 | 4.79 % |

## Orders & Logistics — Business Question

**Question - 7** Does poor delivery performance lead to lower customer satisfaction ?

### SQL Query

1

```sql
WITH Orders_Delivery_Performance AS (
    select
        o.order_sk,
        orv.review_score,
        CAST(
            DATEDIFF(
                DAY,
                o.expected_delivery_date,
                o.delivered_customer_date
            ) As decimal(10,2)
        ) As Delay_Days
    from olist_dwh.gold.dim_orders o
    join olist_dwh.gold.dim_order_reviews orv
        on o.order_sk = orv.order_sk
    where o.order_status = 'DELIVERED'
    AND o.expected_delivery_date IS NOT NULL
    AND o.delivered_customer_date IS NOT NULL
)
```

2

```sql
select
    CASE
        WHEN Delay_Days = 0 THEN 'On Time'
        WHEN Delay_Days < 0 THEN 'Early'
        WHEN Delay_Days BETWEEN 1 AND 3 THEN '1-3 Days Late'
        WHEN Delay_Days BETWEEN 4 AND 7 THEN '4-7 Days Late'
        WHEN Delay_Days BETWEEN 8 AND 14 THEN '8-14 Days Late'
        WHEN Delay_Days BETWEEN 15 AND 30 THEN '15-30 Days Late'
        ELSE '30+ Days Late'
    END As Delay_Range,
    COUNT(order_sk) AS Total_Orders,
    CAST(
        AVG(review_score)
         As decimal(10,2)
    ) As Avg_review_score
from Orders_Delivery_Performance
group by
    CASE
        WHEN Delay_Days = 0 THEN 'On Time'
        WHEN Delay_Days < 0 THEN 'Early'
        WHEN Delay_Days BETWEEN 1 AND 3 THEN '1-3 Days Late'
        WHEN Delay_Days BETWEEN 4 AND 7 THEN '4-7 Days Late'
        WHEN Delay_Days BETWEEN 8 AND 14 THEN '8-14 Days Late'
        WHEN Delay_Days BETWEEN 15 AND 30 THEN '15-30 Days Late'
        ELSE '30+ Days Late'
    END
order by Avg_review_score;
```

| Delay Range | Total Orders | Avg review score |
|---|---|---|
| 15-30 Days Late | 967 | 1 |
| 8-14 Days Late | 1403 | 1 |
| 30+ Days Late | 309 | 2 |
| 4-7 Days Late | 1737 | 2 |
| 1-3 Days Late | 1950 | 3 |
| Early | 88163 | 4 |
| On Time | 1295 | 4 |

## Orders & Logistics — Business Question

**Question - 8** Does delivery performance impact revenue ?

### SQL Query

```sql
WITH Orders_Delivery_Performance AS (
    SELECT
        order_sk,
        order_status,
        CASE
            WHEN delivered_customer_date IS NULL OR expected_delivery_date IS NULL
                THEN 'Unknown'
            WHEN delivered_customer_date > expected_delivery_date
                THEN 'Late Delivery'
            WHEN delivered_customer_date < expected_delivery_date
                THEN 'Early Delivery'
            ELSE 'On Time'
        END AS Delivery_Performance
    FROM olist_dwh.gold.dim_orders
),
Order_Revenue AS (
    SELECT
        order_sk,
        SUM(payment_value) AS Order_Revenue
    FROM olist_dwh.gold.fact_order_payments
    GROUP BY order_sk
)
SELECT
    op.Delivery_Performance,
    COUNT(DISTINCT op.order_sk) AS Total_Orders,
    SUM(orv.Order_Revenue) AS Total_Revenue,
    CAST(
        100.0 * SUM(orv.Order_Revenue)
        / NULLIF(SUM(SUM(orv.Order_Revenue)) OVER (), 0)
        AS DECIMAL(10,2)
    ) AS Revenue_Percentage
FROM Orders_Delivery_Performance op
JOIN Order_Revenue orv
    ON op.order_sk = orv.order_sk
WHERE op.order_status = 'DELIVERED'
GROUP BY op.Delivery_Performance
ORDER BY Total_Revenue DESC;
```

| Delivery Performance | Orders | Revenue | Revenue Percentage |
|---|---|---|---|
| Early Delivery | 88643 | 14303610 | 92.05 |
| Late Delivery | 6519 | 1029892 | 6.63 |
| On Time | 1307 | 205188.94 | 1.32 |
| Unknown | 8 | 1002.49 | 0.01 |

## Business Questions



TOPIC 05 OF 08

# 5. Sales & Revenue Growth

Revenue Trends, Growth Over Time, Sales by Category, Average Order Value

**Proceed to Analytical Questions & Queries** ➡

## Sales & Revenue Growth — Business Question

### Question - 1 How is revenue changing month over month ?

```sql
WITH Monthly_Revenue As (
select
    DATETRUNC(MONTH, o.purchase_date) As Month,
    SUM(op.payment_value) As Current_Month_Revenue,
    LAG(SUM(op.payment_value))
    OVER(order by DATETRUNC(MONTH, o.purchase_date)
    ) As Prev_Month_Revenue
from olist_dwh.gold.fact_order_payments op
left join olist_dwh.gold.dim_orders o
    on op.order_sk = o.order_sk
group by DATETRUNC(MONTH, o.purchase_date)
)
select
    Month,
    Current_Month_Revenue,
    Prev_Month_Revenue,
    Current_Month_Revenue - Prev_Month_Revenue As MOM,
    CAST(
    (Current_Month_Revenue - Prev_Month_Revenue)
    / NULLIF(Prev_Month_Revenue, 0) * 100
    AS decimal(10,2)
    ) As MOM_Percentage,
    CASE
        WHEN Current_Month_Revenue - Prev_Month_Revenue > 0 THEN 'Growing'
        WHEN Current_Month_Revenue - Prev_Month_Revenue < 0 THEN 'Delclining'
        ELSE 'Stable'
    END AS Status
from Monthly_Revenue
where Prev_Month_Revenue IS NOT NULL
order by Month;
```

| Month | Cur Month Revenue | Prev Month Revenue | MOM Change | MOM % | Status |
|---|---|---|---|---|---|
| 2017-01-01 | 127,776.22 | 61.01 | 127,715.21 | 209,334.88 | Growing |
| 2017-02-01 | 289,509.18 | 127,776.22 | 161,732.96 | 126.58 | Growing |
| 2017-03-01 | 418,671 | 289,509.18 | 129,161.82 | 44.61 | Growing |
| 2017-04-01 | 386,739.7 | 418,671 | -31931.3 | -7.63 | Delclining |
| 2017-05-01 | 604,941.48 | 386,739.7 | 218,201.78 | 56.42 | Growing |
| 2017-06-01 | 525,858.59 | 604,941.48 | -79,082.89 | -13.07 | Delclining |
| 2017-07-01 | 626,143.96 | 525,858.59 | 100,285.37 | 19.07 | Growing |
| 2017-08-01 | 666,914.71 | 626,143.96 | 40,770.75 | 6.51 | Growing |
| 2017-09-01 | 682,561.02 | 666,914.71 | 15,646.31 | 2.35 | Growing |
| 2017-10-01 | 764,731.01 | 682,561.02 | 82,169.99 | 12.04 | Growing |
| 2017-11-01 | 1,226,545.16 | 764,731.01 | 461,814.15 | 60.39 | Growing |
| 2017-12-01 | 919,024.11 | 1,226,545.16 | -307,521.05 | -25.07 | Delclining |
| 2018-01-01 | 1,176,530.63 | 919,024.11 | 257,506.52 | 28.02 | Growing |
| 2018-02-01 | 1,091,027.81 | 1,176,530.63 | -85,502.82 | -7.27 | Delclining |
| 2018-03-01 | 1,162,104.55 | 1,091,027.81 | 71,076.74 | 6.51 | Growing |
| 2018-04-01 | 1,104,943.12 | 1,162,104.55 | -57,161.43 | -4.92 | Delclining |
| 2018-05-01 | 1,110,872.8 | 1,104,943.12 | 5,929.68 | 0.54 | Growing |
| 2018-06-01 | 1,012,513.38 | 1,110,872.8 | -98,359.42 | -8.85 | Delclining |
| 2018-07-01 | 1,029,485.86 | 1,012,513.38 | 16,972.48 | 1.68 | Growing |
| 2018-08-01 | 1,020,760.57 | 1,029,485.86 | -8,725.29 | -0.85 | Delclining |
| 2018-09-01 | 1644.89 | 1,020,760.57 | -1,019,115.68 | -99.84 | Delclining |

## Sales & Revenue Growth — Business Question

**Question - 2** Are sales growth and customer growth moving together ?

1

```sql
with QUARTERly_performance as (
    select
        DATETRUNC(QUARTER,o.purchase_date) as QUARTER,
        COUNT(distinct o.order_sk) As Total_Orders,
        SUM(op.payment_value) As Total_Revenue,
        COUNT(distinct c.customer_unique_id) Total_Customers
    from  olist_dwh.gold.dim_orders o
    left join olist_dwh.gold.fact_order_payments op
        on op.order_sk = o.order_sk
    left join olist_dwh.gold.dim_customers c
        on o.customer_sk = c.customer_sk
    group by DATETRUNC(QUARTER,o.purchase_date)
),
QUARTERly_growth As (
    select
        QUARTER,
        Total_Revenue,
        Total_Customers,
        LAG(Total_Revenue) OVER(
          order by QUARTER
        ) As Prev_Revenue,
        LAG(Total_Customers) OVER(
          order by QUARTER
        ) As Prev_Customers
    from QUARTERly_performance
```

2

```sql
select
    QUARTER,
    Total_Revenue,
    Total_Customers,
    Prev_Revenue,
    Prev_Customers,
    CAST(
        (Total_Revenue - Prev_Revenue)
        / NULLIF(Prev_Revenue, 0) * 100
        AS decimal(10,2)
    ) AS Revenue_Growth_Pct,
    CAST(
        (Total_Customers - Prev_Customers) * 100.00
        / NULLIF(Prev_Customers, 0)
        AS decimal(10,2)
    ) AS Customers_Growth_Pct,
    CASE
        WHEN
            (Total_Revenue - Prev_Revenue) > 0
            AND
            (Total_Customers - Prev_Customers) > 0
            THEN 'Both Growing'
        WHEN
            (Total_Revenue - Prev_Revenue) < 0
            AND
            (Total_Customers - Prev_Customers) < 0
            THEN 'Both Declining'
        WHEN
            (Total_Revenue - Prev_Revenue) > 0
            AND
            (Total_Customers - Prev_Customers) < 0
            THEN 'Revenue Up / Customers Down'
        WHEN
            (Total_Revenue - Prev_Revenue) < 0
            AND
            (Total_Customers - Prev_Customers) > 0
            THEN 'Revenue Down / Customers Up'
        ELSE 'Stable'
    END AS Growth_Relationship
from QUARTERly_growth
where Prev_Revenue IS NOT NULL;
```

| Quarter | Total Revenue | Prev Revenue | Revenue Growth Pct | Total Customers | Prev Customers | Customers Growth Pct | Growth Relationship |
|---|---|---|---|---|---|---|---|
| 2016-10-01 | 58729.24 | 527.59 | 11031.61 | 325 | 4 | 8025 | Both Growing |
| 2018-10-01 | 315.54 | 2051891.32 | -99.98 | 4 | 12761 | -99.97 | Both Declining |
| 2017-01-01 | 835956.4 | 58729.24 | 1323.41 | 5249 | 325 | 1515.08 | Both Growing |
| 2017-07-01 | 1975619.69 | 1517539.77 | 30.19 | 12585 | 9322 | 35 | Both Growing |
| 2017-04-01 | 1517539.77 | 835956.4 | 81.53 | 9322 | 5249 | 77.6 | Both Growing |
| 2017-10-01 | 2910300.28 | 1975619.69 | 47.31 | 17713 | 12585 | 40.75 | Both Growing |
| 2018-01-01 | 3429662.99 | 2910300.28 | 17.85 | 21057 | 17713 | 18.88 | Both Growing |
| 2018-04-01 | 3228329.3 | 3429662.99 | -5.87 | 19818 | 21057 | -5.88 | Both Declining |
| 2018-07-01 | 2051891.32 | 3228329.3 | -36.44 | 12761 | 19818 | -35.61 | Both Declining |

## Sales & Revenue Growth — Business Question

**Question - 3** Which months or seasons have the highest sales ?

### SQL Query

```sql
select
    Top 5
    DATETRUNC(MONTH, o.purchase_date) As Month,
    SUM(op.payment_value) As Total_Revenue
from olist_dwh.gold.fact_order_payments op
left join olist_dwh.gold.dim_orders o
    on op.order_sk = o.order_sk
group by DATETRUNC(MONTH, o.purchase_date)
order by Total_Revenue desc;
```

| Quarter | Total Revenue |
|---|---|
| 2017-11-01 | 1,226,545.16 |
| 2018-01-01 | 1,176,530.63 |
| 2018-03-01 | 1,162,104.55 |
| 2018-05-01 | 1,110,872.80 |
| 2018-04-01 | 1,104,943.12 |
| 2018-02-01 | 1,091,027.81 |
| 2018-07-01 | 1,029,485.86 |
| 2018-08-01 | 1,020,760.57 |
| 2018-06-01 | 1,012,513.38 |
| 2017-12-01 | 919,024.11 |

## Sales & Revenue Growth — Business Question

### Question - 4 What are the main drivers behind revenue changes ?

1

```sql
WITH Quarter_Performance AS (
select
    DATETRUNC(QUARTER, o.purchase_date) As Quarter,
    COUNT(distinct o.order_sk) As Total_Orders,
    COUNT(distinct oi.order_item_sk) As Total_Qty,
    SUM(op.payment_value) As Total_Revenue,
    LAG(COUNT(distinct o.order_sk))
    OVER(order by DATETRUNC(Quarter, o.purchase_date)
    ) As Prev_Quarter_Orders,
    LAG(COUNT(distinct oi.order_item_sk))
    OVER(order by DATETRUNC(Quarter, o.purchase_date)
    ) As Prev_Quarter_Qty,
    LAG(SUM(op.payment_value))
    OVER(order by DATETRUNC(Quarter, o.purchase_date)
    ) As Prev_Quarter_Revenue
from olist_dwh.gold.fact_order_payments op
left join olist_dwh.gold.dim_orders o
    on op.order_sk = o.order_sk
left join  olist_dwh.gold.fact_order_items oi
    on o.order_sk = oi.order_sk
group by DATETRUNC(Quarter, o.purchase_date)
),
```

2

```sql
Quarter_QOQ As (
select
    Quarter,
    Total_Orders,
    Total_Qty,
    Total_Revenue,
    Prev_Quarter_Orders,
    Prev_Quarter_Qty,
    Prev_Quarter_Revenue,
    CAST(
    (Total_Orders - Prev_Quarter_Orders) * 100.00
    / NULLIF(Prev_Quarter_Orders,0)
    AS decimal(10,2)
    ) As Orders_QOQ_Pct,
    CAST(
    (Total_Qty - Prev_Quarter_Qty) * 100.00
    / NULLIF(Prev_Quarter_Qty,0)
    AS decimal(10,2)
    ) As Qty_QOQ_Pct,
    CAST(
    (Total_Revenue - Prev_Quarter_Revenue)
    / NULLIF(Prev_Quarter_Revenue,0) * 100
    AS decimal(10,2)
    ) As Revenue_QOQ_Pct
from Quarter_Performance
where Prev_Quarter_Orders IS NOT NULL
)
Select
    Quarter,
    Orders_QOQ_Pct,
    Qty_QOQ_Pct,
    Revenue_QOQ_Pct
from Quarter_QOQ
where  Quarter <> '2018-10-01';
```

| Quarter | Orders QOQ Pct | Qty QOQ Pct | Revenue QOQ Pct |
|---|---|---|---|
| 2016-10-01 | 8025 | 8825 | 12,370.82 |
| 2017-01-01 | 1,519.08 | 1,564.99 | 1,513.12 |
| 2017-04-01 | 77.67 | 78.92 | 83.7 |
| 2017-07-01 | 35.21 | 34.66 | 28.81 |
| 2017-10-01 | 41.19 | 41.48 | 50.29 |
| 2018-01-01 | 18.83 | 19.16 | 16.99 |
| 2018-04-01 | -5.8 | -6.58 | -6.28 |
| 2018-07-01 | -35.84 | -36.07 | -38.32 |



TOPIC 06 OF 08

# 6. Geographic Analysis

Customers by State, Revenue by City, Sellers by State, Delivery by Geographic

**Proceed to Analytical Questions & Queries** ➡

## Geographic Analysis — Business Question

**Question - 1** Which states generate the highest revenue ?

### SQL Query

```sql
select
    Top 10
    COALESCE(l.state, 'N/A') As State,
    SUM(op.payment_value) As Total_Revenue
from olist_dwh.gold.fact_order_payments op
left join  olist_dwh.gold.dim_orders o
    on op.order_sk = o.order_sk
left join  olist_dwh.gold.dim_customers c
    on o.customer_sk = c.customer_sk
left join  olist_dwh.gold.dim_geo_location l
    on c.geo_location_sk = l.geo_location_sk
group by COALESCE(l.state, 'N/A')
order by Total_Revenue desc;
```

| State | Total Revenue |
|---|---|
| SAO PAULO | 6,693,100.71 |
| RIO DE JANEIRO | 2,085,766.18 |
| MINAS GERAIS | 1,867,835.70 |
| RIO GRANDE DO SUL | 882,545.07 |
| PARANA | 805,367.63 |
| SANTA CATARINA | 579,813.45 |
| BAHIA | 562,995.11 |
| GOIAS | 323,854.11 |
| ESPIRITO SANTO | 321,326.28 |
| DISTRITO FEDERAL | 301,398.04 |

## Geographic Analysis — Business Question

**Question - 2** Which states have high order volume but low revenue ?

### SQL Query

```sql
with States_Performance as (
select
    COALESCE(l.state, 'N/A') As State,
    COUNT(DISTINCT c.customer_unique_id) As Customers,
    COUNT(distinct o.order_sk) As Total_Orders,
    SUM(op.payment_value) As Total_Revenue,
    CAST(
    SUM(op.payment_value) / COUNT(distinct o.order_sk)
    As decimal(10,2)
    ) As AOV
from olist_dwh.gold.dim_orders o
left join olist_dwh.gold.fact_order_payments op
    on op.order_sk = o.order_sk
left join olist_dwh.gold.dim_customers c
    on o.customer_sk = c.customer_sk
left join olist_dwh.gold.dim_geo_location l
    on c.geo_location_sk = l.geo_location_sk
group by COALESCE(l.state, 'N/A')
)
select
    *
from States_Performance
where
    Total_Orders > 1000
    AND Total_Revenue < 500000
order by Total_Orders DESC;
```

| State | Customers | Total Orders | Total Revenue | AOV |
|---|---|---|---|---|
| ESPIRITO SANTO | 1,958 | 2,026 | 321,326.28 | 158.6 |
| GOIAS | 1,944 | 2,011 | 323,854.11 | 161.04 |
| DISTRITO FEDERAL | 1,904 | 1,964 | 301,398.04 | 153.46 |
| PERNAMBUCO | 1,605 | 1,648 | 255,903.35 | 155.28 |
| CEARA | 1,310 | 1,332 | 207,553 | 155.82 |

## Geographic Analysis — Business Question

**Question - 3** Which states have the highest number of customers ?

### SQL Query

```
select
    Top 10
    l.city,
    COUNT(DISTINCT c.customer_unique_id) As Customers
from olist_dwh.gold.dim_customers c
join olist_dwh.gold.dim_geo_location l
    on c.geo_location_sk = l.geo_location_sk
group by l.city
order by Customers DESC;
```

| City | Customers |
|---|---|
| SAO PAULO | 14982 |
| RIO DE JANEIRO | 6620 |
| BELO HORIZONTE | 2672 |
| BRASILIA | 1898 |
| CURITIBA | 1465 |
| CAMPINAS | 1398 |
| PORTO ALEGRE | 1326 |
| SALVADOR | 1205 |
| GUARULHOS | 1153 |
| SAO BERNARDO DO CAMPO | 908 |

## Geographic Analysis — Business Question

**Question - 4** Which regions have the longest delivery times ?

### SQL Query

```sql
select
    Top 5
    l.state,
    CAST(
        AVG(
            DATEDIFF(
                DAY,
                o.purchase_date,
                o.delivered_customer_date
            ) * 1.0
        )
        AS Decimal(10,2)
    ) As Avg_Delivery_Days
from olist_dwh.gold.dim_orders o
join olist_dwh.gold.dim_customers c
    on o.customer_sk = c.customer_sk
join olist_dwh.gold.dim_geo_location l
    on c.geo_location_sk = l.geo_location_sk
group by l.state
order by Avg_Delivery_Days DESC;
```

| State | Avg Delivery Days |
|---|---|
| AMAPA | 16.18 |
| AMAZONAS | 14.98 |
| RORAIMA | 14.41 |
| TOCANTINS | 14.23 |
| PARAIBA | 14.02 |
| Unknown | 13.93 |
| MATO GROSSO DO SUL | 12.98 |
| PERNAMBUCO | 12.84 |
| RIO GRANDE DO NORTE | 12.76 |
| RIO GRANDE DO SUL | 12.75 |

## Geographic Analysis — Business Question

**Question - 5** Which states have the highest cancellation rates ?

### SQL Query

```sql
select
    Top 10
    l.state,
    COUNT(o.order_sk) As Total_Orders,
    COUNT(
        CASE
            WHEN o.order_status = 'CANCELED' THEN o.order_sk
        END ) As Canceled_Orders,
    CAST(
        COUNT(CASE
                WHEN o.order_status = 'CANCELED' THEN o.order_sk
            END ) * 100.00
        / COUNT(o.order_sk)
          As decimal(10,2)
    ) As Canceled_Orders_Rate
from olist_dwh.gold.dim_orders o
join olist_dwh.gold.dim_customers c
    on o.customer_sk = c.customer_sk
join olist_dwh.gold.dim_geo_location l
    on c.geo_location_sk = l.geo_location_sk
group by l.state
order by Canceled_Orders_Rate DESC;
```

| State | Total Orders | Canceled Orders | Canceled Orders Rate |
|---|---|---|---|
| AMAPA | 68 | 2 | 2.94 % |
| ACRE | 81 | 2 | 2.47 % |
| MARANHAO | 743 | 10 | 1.35 % |
| MATO GROSSO DO SUL | 715 | 9 | 1.26 % |
| RIO GRANDE DO NORTE | 483 | 6 | 1.24 % |
| PARAIBA | 534 | 5 | 0.94 % |
| PARANA | 5032 | 45 | 0.89 % |
| SERGIPE | 349 | 3 | 0.86 % |
| PARA | 971 | 8 | 0.82 % |
| SANTA CATARINA | 3637 | 29 | 0.8 % |

## Geographic Analysis — Business Question

### Question - 6 Are there underserved regions with high demand ?

1

```sql
WITH Orders_Performance As (
    select
        o.order_sk,--
        l.state,--
        o.order_status,--
        DATEDIFF(DAY,
            o.purchase_date,
            o.delivered_customer_date
        ) As Delivery_Days,
        DATEDIFF(DAY,
            o.expected_delivery_date,
            o.delivered_customer_date
        ) As Delayed_Delivery_Days,
        CASE
            WHEN o.order_status = 'CANCELED' THEN 'Canceled'
            WHEN o.delivered_customer_date IS NULL OR o.expected_delivery_date IS NULL
                THEN 'Unknown'
            WHEN o.order_status = 'DELIVERED'
                AND o.delivered_customer_date > o.expected_delivery_date
                THEN 'Late Delivered'
            WHEN o.order_status = 'DELIVERED'
                AND o.delivered_customer_date < o.expected_delivery_date
                THEN 'Early Delivered'
            ELSE 'On Time'
        END As Order_Performance,
        orv.review_score
    from olist_dwh.gold.dim_orders o
    join olist_dwh.gold.dim_customers c
        on o.customer_sk = c.customer_sk
    join olist_dwh.gold.dim_geo_location l
        on c.geo_location_sk = l.geo_location_sk
    left join olist_dwh.gold.dim_order_reviews orv
        on o.order_sk = orv.order_sk
    where o.order_status IN ('DELIVERED', 'CANCELED')
),
```

2

```sql
Orders_Performance_by_State As (
    select
        state,
        COUNT(order_sk) As Total_Orders,

        --******* Delivered Orders - Delivery Rate  *******
        CAST(
            SUM(CASE
                    WHEN order_status = 'DELIVERED' THEN 1
                    ELSE 0
                END ) * 100.00
            / COUNT(order_sk)
              As decimal(10,2)
        ) As Delivered_Orders_Rate,
        ------------------------------------------------
        -- ** Order_Performance breakedown
        ------------------------------------------------
        CAST(
            SUM(
                CASE
                    WHEN Order_Performance = 'Late Delivered' THEN 1
                    ELSE 0
                END) * 100.00
            / SUM(
                CASE
                    WHEN order_status = 'DELIVERED' THEN 1
                    ELSE 0
                END)
             As decimal(10,2)
        ) As Late_Delivered_Rate,

        CAST(
            SUM(
                CASE
                    WHEN Order_Performance = 'Early Delivered' THEN 1
                    ELSE 0
                END) * 100.00
            / SUM(
                CASE
                    WHEN order_status = 'DELIVERED' THEN 1
                    ELSE 0
                END)
             As decimal(10,2)
        ) As Early_Delivered_Rate,
```

3

```sql
        CAST(
            SUM(
                CASE
                    WHEN Order_Performance = 'On Time' THEN 1
                    ELSE 0
                END) * 100.00
            / SUM(
                CASE
                    WHEN order_status = 'DELIVERED' THEN 1
                    ELSE 0
                END)
             As decimal(10,2)
        ) As OnTime_Delivered_Rate,
        ------------------------------------------------
        -- ** Canceled Orders - Delivery Rate
        ------------------------------------------------
        CAST(
            SUM(CASE
                    WHEN order_status = 'CANCELED' THEN 1
                    ELSE 0
                END ) * 100.00
            / COUNT(order_sk)
              As decimal(10,2)
        ) As Canceled_Orders_Rate,
        ------------------------------------------------
        -- ** Avg_review_score
        ------------------------------------------------
        CAST(
            AVG(review_score * 1.0)
              As decimal(10,2)
        ) As Avg_review_score

    from Orders_Performance
    group by state
)
select *
from Orders_Performance_by_State
where Total_Orders > 500 AND Late_Delivered_Rate < 6.5
order by Total_Orders DESC;
```

| State | Total Orders | Delivered Orders Rate | Late Delivered Rate | Early Delivered Rate | OnTime Delivered Rate | Canceled Orders Rate | Avg review score |
|---|---|---|---|---|---|---|---|
| GOIAS | 1970 | 99.59 | 6.27 | 92.41 | 1.33 | 0.41 | 4.15 |
| CEARA | 1295 | 99.38 | 5.91 | 92.46 | 1.63 | 0.62 | 4.12 |
| PARA | 950 | 99.16 | 5.94 | 93.42 | 0.64 | 0.84 | 4.23 |
| MARANHAO | 724 | 98.62 | 5.6 | 92.72 | 1.68 | 1.38 | 4.09 |

## Business Questions



TOPIC 07 OF 08

# 7. Payments Analytics

Payments Methods, Payment Value, Payment Mix, Installments, Orders by Methods

**Proceed to Analytical Questions & Queries** ➡

## Question - 1 Which payment methods are most popular ?

### SQL Query

```sql
select
    payment_type,
    COUNT(*) As Payment_Count,
    SUM(payment_value) As Total_Payment_Value
from olist_dwh.gold.fact_order_payments
group by payment_type
order by Total_Payment_Value desc;
```

| | Payment Type | Payment Count | Total Payment Value |
|---|---|---|---|
| 1 | Credit Card | 76795 | 12,542,084.19 |
| 2 | Boleto | 19784 | 28,693,61.27 |
| 3 | Voucher | 5775 | 379,436.87 |
| 4 | Debit Card | 1529 | 217,989.79 |
| 5 | Not Defined | 3 | 0 |

## Question - 2 How has payment method usage changed over time ?

**SQL Query**

1

```sql
WITH Payments_Types_Performance As(
select
    YEAR(o.purchase_date) As Year,
    op.payment_type,
    COUNT(op.payment_type) As Payment_Count,
    SUM(op.payment_value) As Total_Payment_Value
from olist_dwh.gold.fact_order_payments op
join olist_dwh.gold.dim_orders o
    on op.order_sk = o.order_sk
group by YEAR(o.purchase_date), op.payment_type
),
```

2

```sql
Performance_YOY as (
select
    Year,
    payment_type,
    Payment_Count,
    COALESCE(
        LAG(Payment_Count) OVER(
            PARTITION BY payment_type
            ORDER BY Year ) ,0
    ) As Prev_Payment_Count,
    COALESCE(
        Payment_Count
        - LAG(Payment_Count) OVER(
            PARTITION BY payment_type
            ORDER BY Year) , 0
    ) As Payment_Count_Change,
    Total_Payment_Value
from Payments_Types_Performance
)
select *
from Performance_YOY
where Year = 2018;
```

| Year | payment type | Payment Count | Prev Payment Count | Payment Count Change | Total Payment Value |
|---|---|---|---|---|---|
| 2018 | Credit Card | 41,665 | 34,884 | 6,781 | 6,830,833.56 |
| 2018 | Boleto | 10,779 | 8,929 | 1,850 | 1,558,591.84 |
| 2018 | Voucher | 3,158 | 2,605 | 553 | 196,549.67 |
| 2018 | Debit Card | 832 | 693 | 139 | 124,224.08 |
| 2018 | Not Defined | 2 | 1 | 1 | 0 |

## Question - 3 Which state prefer which payment method ?

```sql
WITH Orders_per_Payment_Type As (
select
    o.order_sk,
    l.state,
    op.payment_type
from olist_dwh.gold.fact_order_payments op
join olist_dwh.gold.dim_orders o
    on op.order_sk = o.order_sk
join olist_dwh.gold.dim_customers c
    on o.customer_sk = c.customer_sk
join olist_dwh.gold.dim_geo_location l
    on c.geo_location_sk = l.geo_location_sk
),
Payment_Count As(
select
    state,
    payment_type,
    COUNT(*) As Payment_Count,
    ROW_NUMBER() OVER(
        PARTITION BY state
        ORDER BY COUNT(*) DESC
    ) As Payment_Rank
from Orders_per_Payment_Type
group by state, payment_type
)
select *
from Payment_Count
where Payment_Rank = 1
order by state;
```

| State | Payment Type | Payment Count |
|---|---|---|
| ACRE | Credit Card | 64 |
| ALAGOAS | Credit Card | 310 |
| AMAPA | Credit Card | 50 |
| AMAZONAS | Credit Card | 112 |
| BAHIA | Credit Card | 2,613 |
| CEARA | Credit Card | 1,046 |
| DISTRITO FEDERAL | Credit Card | 1,525 |
| ESPIRITO SANTO | Credit Card | 1,562 |
| GOIAS | Credit Card | 1,529 |
| MARANHAO | Credit Card | 585 |
| MATO GROSSO | Credit Card | 685 |
| MATO GROSSO DO SUL | Credit Card | 548 |
| MINAS GERAIS | Credit Card | 8,993 |
| PARA | Credit Card | 753 |
| PARAIBA | Credit Card | 405 |
| PARANA | Credit Card | 3,873 |
| PERNAMBUCO | Credit Card | 1,242 |
| PIAUI | Credit Card | 372 |
| RIO DE JANEIRO | Credit Card | 9,860 |
| RIO GRANDE DO NORTE | Credit Card | 385 |
| RIO GRANDE DO SUL | Credit Card | 4,179 |
| RONDONIA | Credit Card | 200 |
| RORAIMA | Credit Card | 37 |
| SANTA CATARINA | Credit Card | 2,845 |
| SAO PAULO | Credit Card | 32,306 |
| SERGIPE | Credit Card | 266 |
| TOCANTINS | Credit Card | 223 |
| Unknown | Credit Card | 227 |

## Question - 4 What is the average payment value by payment method ? And Which payment methods are associated with larger purchases ?

```sql
select
    payment_type,
    COUNT(*) As Payment_Count,
    SUM(payment_value) As Total_Payment_Value,
    CAST(
        SUM(payment_value)
        / NULLIF(COUNT(*), 0)
          As decimal(10,2)
    ) As Avg_Payment_Method
from olist_dwh.gold.fact_order_payments
group by payment_type
order by Avg_Payment_Method desc;
```

| Payment Type | Payment Count | Total Payment Value | Avg Payment Method |
|---|---|---|---|
| Credit Card | 76,795 | 12,542,084.19 | 163.32 |
| Boleto | 19,784 | 2,869,361.27 | 145.03 |
| Debit Card | 1,529 | 217,989.79 | 142.57 |
| Voucher | 5,775 | 379,436.87 | 65.7 |
| Not Defined | 3 | 0 | 0 |

## Question - 5 How many installments are customers using ?

```sql
select
    Top 5
    payment_installments,
    COUNT(*) As Payment_Count
from olist_dwh.gold.fact_order_payments
group by payment_installments
order by Payment_Count DESC;
```

| payment installments | Payment Count |
|---|---|
| 1 | 52,546 |
| 2 | 12,413 |
| 3 | 10,461 |
| 4 | 7,098 |
| 10 | 5,328 |

## Question - 6 Does installments usage correlate with higher order values ?

### SQL Query

```sql
WITH Orders_Payment_Summary AS (
    SELECT
        order_sk,
        MAX(payment_installments) AS payment_installments,
        SUM(payment_value) AS Order_Payment_Value,
        COUNT(*) AS Payment_Records
    FROM olist_dwh.gold.fact_order_payments
    GROUP BY order_sk
)
SELECT
    TOP 10
    payment_installments,
    COUNT(*) AS Orders_Count,
    CAST(
        AVG(Order_Payment_Value) AS DECIMAL(12,2)
    ) AS Avg_Order_Payment_Value,
    CAST(
        SUM(Order_Payment_Value) AS DECIMAL(14,2)
    ) AS Total_Payment_Value
FROM Orders_Payment_Summary
WHERE Payment_Records = 1
GROUP BY payment_installments
ORDER BY Avg_Order_Payment_Value DESC;
```

| payment installments | Orders Count | Avg Order Payment Value | Total Payment Value |
|---|---|---|---|
| 24 | 16 | 627.93 | 100,46.91 |
| 20 | 17 | 615.8 | 10,468.63 |
| 18 | 26 | 492.84 | 12,813.85 |
| 15 | 72 | 436.65 | 31,438.54 |
| 10 | 5,199 | 414.37 | 2,154,301.62 |
| 12 | 129 | 324.34 | 41,840.02 |
| 8 | 4,189 | 307.65 | 1,288,766.28 |
| 16 | 5 | 292.69 | 1,463.47 |
| 21 | 3 | 243.7 | 731.1 |
| 23 | 1 | 236.48 | 236.48 |



TOPIC 08 OF 08

# 8. Reviews & Customer Satisfaction

Review Scores, Seller Reviews, Operational Patterns, Reviews by Category, Delivery vs Reviews

**Proceed to Analytical Questions & Queries** ➡

## Reviews & Customer Satisfaction — Business Question

### Question - 1 Does late delivery reduce review scores ?

**SQL Query**

```sql
WITH Orders_Status As (
    select
        o.order_sk,
        CASE
            WHEN o.delivered_customer_date > o.expected_delivery_date THEN 'Late Delivery'
            WHEN o.delivered_customer_date < o.expected_delivery_date THEN 'Early Delivery'
            ELSE 'On Time'
        END AS Delivery_Status,
        orv.review_score
    from olist_dwh.gold.dim_orders o
    left join olist_dwh.gold.dim_order_reviews orv
        on o.order_sk = orv.order_sk
)
Select
    Delivery_Status,
    COUNT(review_score) As Reviewed_Orders,
    CAST(
        AVG(review_score * 1.0)
          As decimal(10,2)
    ) As Avg_review_Score
from Orders_Status
group by Delivery_Status
order by Avg_review_Score DESC;
```

| Delivery Status | Reviewed Orders | Avg Review Score |
|---|---|---|
| Early Delivery | 88,168 | 4.29 |

| Delivery Status | Reviewed Orders | Avg Review Score |
|---|---|---|
| On Time | 4,138 | 2.46 |

| Delivery Status | Reviewed Orders | Avg Review Score |
|---|---|---|
| Late Delivery | 6,367 | 2.27 |

## Reviews & Customer Satisfaction — Business Question

**Question - 2** Which sellers have the highest ratings ?

### SQL Query

```sql
with Orders_Review As (
    select
        o.order_sk,
        orv.review_score
    from olist_dwh.gold.dim_orders o
    left join olist_dwh.gold.dim_order_reviews orv
        on o.order_sk = orv.order_sk
),
Orders_by_Seller As (
    select Distinct
        order_sk,
        seller_sk
    from olist_dwh.gold.fact_order_items
)
select
    TOP 5
    os.seller_sk,
    COUNT(orv.review_score) As Reviewed_Orders,
    CAST(
        AVG(orv.review_score * 1.0)
        As decimal(10,2)
    ) As Avg_review_score
from Orders_Review orv
join Orders_by_Seller os
    on orv.order_sk = os.order_sk
group by os.seller_sk
having COUNT(orv.review_score) > 100
order by Avg_review_score desc;
```

| Seller SK | Reviewed Orders | Avg Review Score |
|---|---|---|
| 2389 | 123 | 4.33 |
| 308 | 163 | 4.32 |
| 820 | 108 | 4.31 |
| 665 | 116 | 4.28 |
| 508 | 185 | 4.25 |

## Reviews & Customer Satisfaction — Business Question

**Question - 3** Which categories receive the best reviews ?

### SQL Query

```sql
with Order_Category As (
select
    DISTINCT
    oi.order_sk,
    pc.product_category_name_engilsh
from olist_dwh.gold.fact_order_items oi
join olist_dwh.gold.dim_products p
    on oi.product_sk = p.product_sk
join olist_dwh.gold.dim_product_categories pc
    on p.product_category_id = pc.product_category_id
)
select
    Top 10
    oc.product_category_name_engilsh,
    CAST(
        AVG(orv.review_score * 1.0)
           As decimal(10,2)
    ) As Avg_Score
from Order_Category oc
join olist_dwh.gold.dim_order_reviews orv
    on oc.order_sk = orv.order_sk
group by oc.product_category_name_engilsh
order by Avg_Score DESC;
```

| Product Category Name | Avg Score |
|---|---|
| Security And Services | 5 |
| Tablets Printing Image | 4.31 |
| Books Imported | 4.3 |
| Fashion Childrens Clothes | 4.25 |
| Party Supplies | 4.23 |
| Industry Commerce And Business | 4.22 |
| Food | 4.22 |
| Costruction Tools Garden | 4.2 |
| Christmas Supplies | 4.18 |
| Signaling And Security | 4.17 |

## Reviews & Customer Satisfaction — Business Question

**Question - 4** Are expensive products rated differently from cheaper products ?

### SQL Query

```sql
WITH Product_Price_Range As (
select DISTINCT
    order_sk,
    product_sk,
    price,
    CASE
        WHEN price BETWEEN 0 AND 50 THEN '0-50'
        WHEN price BETWEEN 51 AND 100 THEN '51-100'
        WHEN price BETWEEN 101 AND 500 THEN '101-500'
        WHEN price BETWEEN 501 AND 1000 THEN '501-1000'
        ELSE 'Above 1000'
    END Price_Range
from olist_dwh.gold.fact_order_items
),
Orders_Review As (
select
    o.order_Sk,
    orv.review_score
from olist_dwh.gold.dim_orders o
left join olist_dwh.gold.dim_order_reviews orv
    on o.order_sk = orv.order_sk
)
select
    ppr.Price_Range,
    COUNT(ppr.product_sk) As Total_Order_Items,
    CAST(
        AVG(orv.review_score * 1.0)
        As decimal(10,2)
    ) Avg_Review_Score
from Product_Price_Range ppr
join Orders_Review orv
    on ppr.order_sk = orv.order_sk
group by ppr.Price_Range;
```

| Price Range | Total Order Items | Avg Review Score |
|---|---|---|
| 0-50 | 34,409 | 4.09 |
| 101-500 | 34,738 | 4.09 |
| 501-1000 | 2,300 | 4.05 |
| 51-100 | 29,927 | 4.08 |
| Above 1000 | 1,051 | 4.1 |

## Reviews & Customer Satisfaction — Business Question

**Question - 5** Which states have lower customer satisfaction ?

### SQL Query

```sql
WITH orders_Review_by_State As (
    select
        o.order_sk,
        l.state,
        orv.review_score
    from olist_dwh.gold.dim_orders o
    left join olist_dwh.gold.dim_order_reviews orv
        on o.order_sk = orv.order_sk
    left join olist_dwh.gold.dim_customers c
        on o.customer_sk = c.customer_sk
    left join olist_dwh.gold.dim_geo_location l
        on c.geo_location_sk = l.geo_location_sk
)
select
    Top 5
    state,
    COUNT(review_score) As Reviewed_Orders,
    CAST(
        AVG(review_score * 1.0)
          As decimal(10,2)
    ) Avg_Review_Score
from orders_Review_by_State
group by state
having COUNT(review_score) >= 100
order by Avg_Review_Score;
```

| State | Reviewed Orders | Avg Review Score |
|---|---|---|
| AMAZONAS | 145 | 3.98 |
| PARAIBA | 530 | 3.98 |
| RIO GRANDE DO NORTE | 482 | 4 |
| MARANHAO | 740 | 4.02 |
| PARANA | 4989 | 4.05 |

## Reviews & Customer Satisfaction — Business Question

**Question - 6** What are the main operational factors associated with poor reviews ?

### (1)- Delivery Delay

```sql
WITH Orders_by_Delivery_Performance As (
    select
        o.order_sk,
        orv.review_score,
        CASE
            WHEN o.delivered_customer_date IS NULL OR o.expected_delivery_date IS NULL
                THEN 'UnKnown'
            WHEN o.delivered_customer_date > o.expected_delivery_date THEN 'Late Delivery'
            WHEN o.delivered_customer_date < o.expected_delivery_date THEN 'Early Delivery'
            ELSE 'On Time'
        END As Delivery_Performance
    from olist_dwh.gold.dim_orders o
    left join olist_dwh.gold.dim_order_reviews orv
        on o.order_sk = orv.order_sk
    where o.order_status = 'DELIVERED'
)
select
    Delivery_Performance,
    COUNT(review_score) AS Reviewed_Orders,
    CAST(
        AVG(review_score * 1.0)
          As decimal(10,2)
    ) Avg_Review_Score,
    CAST(
        SUM(
            CASE
                WHEN review_score <= 2 THEN 1
                ELSE 0
            END) * 100.00
            / NULLIF(COUNT(review_score) ,0)
        AS decimal(10,2)
    ) As Poor_Review_Rate
from Orders_by_Delivery_Performance
group by Delivery_Performance
order by Poor_Review_Rate DESC;
```

| Delivery Performance | Reviewed Orders | Avg Review Score | Poor Review Rate |
|---|---|---|---|
| Late Delivery | 6,366 | 2.27 | 62.5 |
| Unknown | 8 | 4.5 | 12.5 |
| On Time | 1,295 | 4.04 | 12.43 |
| Early Delivery | 88,163 | 4.29 | 9.22 |

## (2)- Delivery Duration

### SQL Query

```sql
WITH Orders_by_Delivery_Duration As (
    select
        o.order_sk,
        DATEDIFF(
            DAY, o.purchase_date, o.delivered_customer_date
        ) As Delivery_Duration,
        orv.review_score
    from olist_dwh.gold.dim_orders o
    left join olist_dwh.gold.dim_order_reviews orv
        on o.order_sk = orv.order_sk
    where o.order_status = 'DELIVERED'
)
select
    CASE
        WHEN Delivery_Duration BETWEEN 1 AND 3 THEN '1-3 Days'
        WHEN Delivery_Duration BETWEEN 4 AND 7 THEN '4-7 Days'
        WHEN Delivery_Duration BETWEEN 8 AND 14 THEN '8-14 Days'
        WHEN Delivery_Duration BETWEEN 15 AND 30 THEN '15-30 Days'
        ELSE 'Above 30 Days'
    END As Delivery_Duration_Range,
    COUNT(review_score) As Reviewed_Orders,
    CAST(
        AVG(review_score * 1.0)
          As decimal(10,2)
    ) Avg_Review_Score,
    CAST(
        SUM(
            CASE
                WHEN review_score <= 2 THEN 1
                ELSE 0
            END) * 100.00
            / NULLIF(COUNT(review_score) ,0)
        AS decimal(10,2)
    ) As Poor_Review_Rate
from Orders_by_Delivery_Duration
group by
    CASE
        WHEN Delivery_Duration BETWEEN 1 AND 3 THEN '1-3 Days'
        WHEN Delivery_Duration BETWEEN 4 AND 7 THEN '4-7 Days'
        WHEN Delivery_Duration BETWEEN 8 AND 14 THEN '8-14 Days'
        WHEN Delivery_Duration BETWEEN 15 AND 30 THEN '15-30 Days'
        ELSE 'Above 30 Days'
    END
order by MIN(Delivery_Duration);
```

| Delivery Duration Range | Reviewed Orders | Avg Review Score | Poor Review Rate |
|---|---|---|---|
| Above 30 Days | 4,190 | 2.21 | 64.37 |
| 1-3 Days | 6,913 | 4.46 | 7.03 |
| 4-7 Days | 23,621 | 4.4 | 7.69 |
| 8-14 Days | 37,785 | 4.3 | 9.1 |
| 15-30 Days | 23,323 | 3.94 | 16.43 |

## Reviews & Customer Satisfaction — Business Question

# (3)- Sellers Performance

**SQL Query**

1

```sql
with Orders_Reviews As (
    select DISTINCT
        o.order_sk,
        orv.review_score,
        oi.seller_sk
    from olist_dwh.gold.dim_orders o
    left join olist_dwh.gold.dim_order_reviews orv
        on o.order_sk = orv.order_sk
    left join olist_dwh.gold.fact_order_items oi
        on o.order_sk = oi.order_sk
    where o.order_status = 'DELIVERED'
),
Orders_Range As (
    select
        seller_sk,
        COUNT(order_sk) As Reviewed_Orders,
        CAST(
            AVG(review_score * 1.0)
              As decimal(10,2)
        ) As Avg_Review_Score,
        SUM(
            CASE
                WHEN review_score <= 2 THEN 1
                ELSE 0
            END
        ) As Poor_Review,
        CASE
            WHEN COUNT(review_score) BETWEEN 1 AND 20 THEN '1-20 Orders'
            WHEN COUNT(review_score) BETWEEN 21 AND 50 THEN '21-50 Orders'
            WHEN COUNT(review_score) BETWEEN 51 AND 100 THEN '51-100 Orders'
            WHEN COUNT(review_score) BETWEEN 101 AND 500 THEN '101-500 Orders'
            WHEN COUNT(review_score) BETWEEN 501 AND 1000 THEN '501-1000 Orders'
            ELSE 'Above 1000 Orders'
        END Orders_Range
    from Orders_Reviews
    group by seller_sk
)
```

2

```sql
select
    Orders_Range,
    COUNT(seller_sk) As Sellers,
    CAST(
        AVG(Avg_Review_Score * 1.0)
          As decimal(10,2)
    ) As Avg_Review_Score,
    CAST(
        SUM(Poor_Review) * 100.0
        / NULLIF(SUM(Reviewed_Orders), 0)
        AS DECIMAL(10,2)
    ) AS Poor_Review_Rate
from Orders_Range
group by Orders_Range
order by Poor_Review_Rate;
```

| Orders Range | Sellers | Avg Review Score | Poor Review Rate |
|---|---|---|---|
| 1-20 Orders | 2,282 | 4.16 | 12.44 |
| 51-100 Orders | 203 | 4.15 | 12.53 |
| 21-50 Orders | 377 | 4.16 | 12.75 |
| 101-500 Orders | 179 | 4.15 | 12.79 |
| Above 1000 Orders | 17 | 4.15 | 12.81 |
| 501-1000 Orders | 13 | 4.14 | 13.12 |

## (4)- Freight Cost

### SQL Query

```sql
WITH Orders_Delivery_Value As (
    select DISTINCT
        oi.order_sk,
        oi.product_sk,
        oi.freight_value,
        orv.review_score,
        CASE
            WHEN oi.freight_value BETWEEN 0 AND 10 THEN '0-10 $'
            WHEN oi.freight_value BETWEEN 11 AND 20 THEN '11-20 $'
            WHEN oi.freight_value BETWEEN 21 AND 50 THEN '21-50 $'
            WHEN oi.freight_value BETWEEN 51 AND 100 THEN '51-100 $'
            ELSE 'Above 100 $'
        END As freight_value_Range
    from olist_dwh.gold.fact_order_items oi
    left join olist_dwh.gold.dim_order_reviews orv
        on oi.order_sk = orv.order_sk
)
select
    freight_value_Range,
    COUNT(product_sk) As Products,
    CAST(
        AVG(review_score * 1.0)
          As decimal(10,2)
    ) As Avg_Review_Score,
    CAST(
        SUM(
            CASE
                WHEN review_score <= 2 THEN 1
                ELSE 0
            END) * 100.00
            / NULLIF(COUNT(review_score) ,0)
        AS decimal(10,2)
    ) As Poor_Review_Rate
from Orders_Delivery_Value
group by freight_value_Range
order by Poor_Review_Rate DESC;
```

| Freight Value Range | Products | Avg Review Score | Poor Review Rate |
|---|---|---|---|
| 51-100 $ | 3438 | 4.08 | 15.36 |
| 21-50 $ | 22166 | 4.07 | 14.95 |
| 11-20 $ | 58239 | 4.09 | 14.6 |
| Above 100 $ | 4484 | 4.08 | 14.57 |
| 0-10 $ | 14098 | 4.1 | 14.36 |

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# Key Executive Findings Summary

## Business Growth & Market Performance：

**01** **Revenue Growth & Customer Acquisition**

Revenue increased from **R$7.14M in 2017 to R$8.64M in 2018**, while the number of customers placing orders grew from **44,358 to 52,984**. This indicates business growth alongside an expanding customer base. *Note: 2018 data ends on October 17, 2018.*

**02** **Geographic Revenue Concentration**

**São Paulo (41.81%), Rio de Janeiro (13.03%), and Minas Gerais (11.67%)** collectively contributed approximately **66.5% of total revenue**. Maintaining performance in these key markets while exploring growth opportunities in other regions could help reduce geographic concentration.

**03** **Credit Cards Lead Payment Value**

Credit card payments generated approximately **R$12.54M**, compared with **R$2.87M through Boleto**. Credit cards represent the largest payment channel by transaction value, making payment performance and reliability important areas to monitor.

**04** **Product Categories Show Different Sales Patterns**

**Health & Beauty (R$1.44M)** led the analyzed categories by revenue, followed by **Watches & Gifts (R$1.31M)** and **Bed Bath & Table (R$1.24M)**. Since revenue and order volume vary across categories, performance should be evaluated using multiple metrics rather than revenue alone.

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# Key Executive Findings Summary

## Customer Experience & Growth Opportunities：

05 **Delivery Performance & Customer Satisfaction**

Late deliveries received an average review score of **2.27**, compared with **4.29 for early deliveries**. This suggests a potential relationship between delivery performance and customer satisfaction, highlighting the importance of reliable fulfillment.

06 **Late Deliveries as an Operational Priority**

Late deliveries accounted for approximately **6.76% of delivered orders** in the analyzed dataset. Identifying the regions, sellers, and product categories with higher delay rates could help prioritize operational improvements.

07 **Repeat Customers Present a Retention Opportunity**

Repeat customers represented approximately **3.14% of customers**, compared with **96.86% who purchased once**. Average revenue per repeat customer was **R$343.29**, versus **R$160.38 for one-time customers**. This suggests an opportunity to investigate retention strategies and encourage repeat purchases.

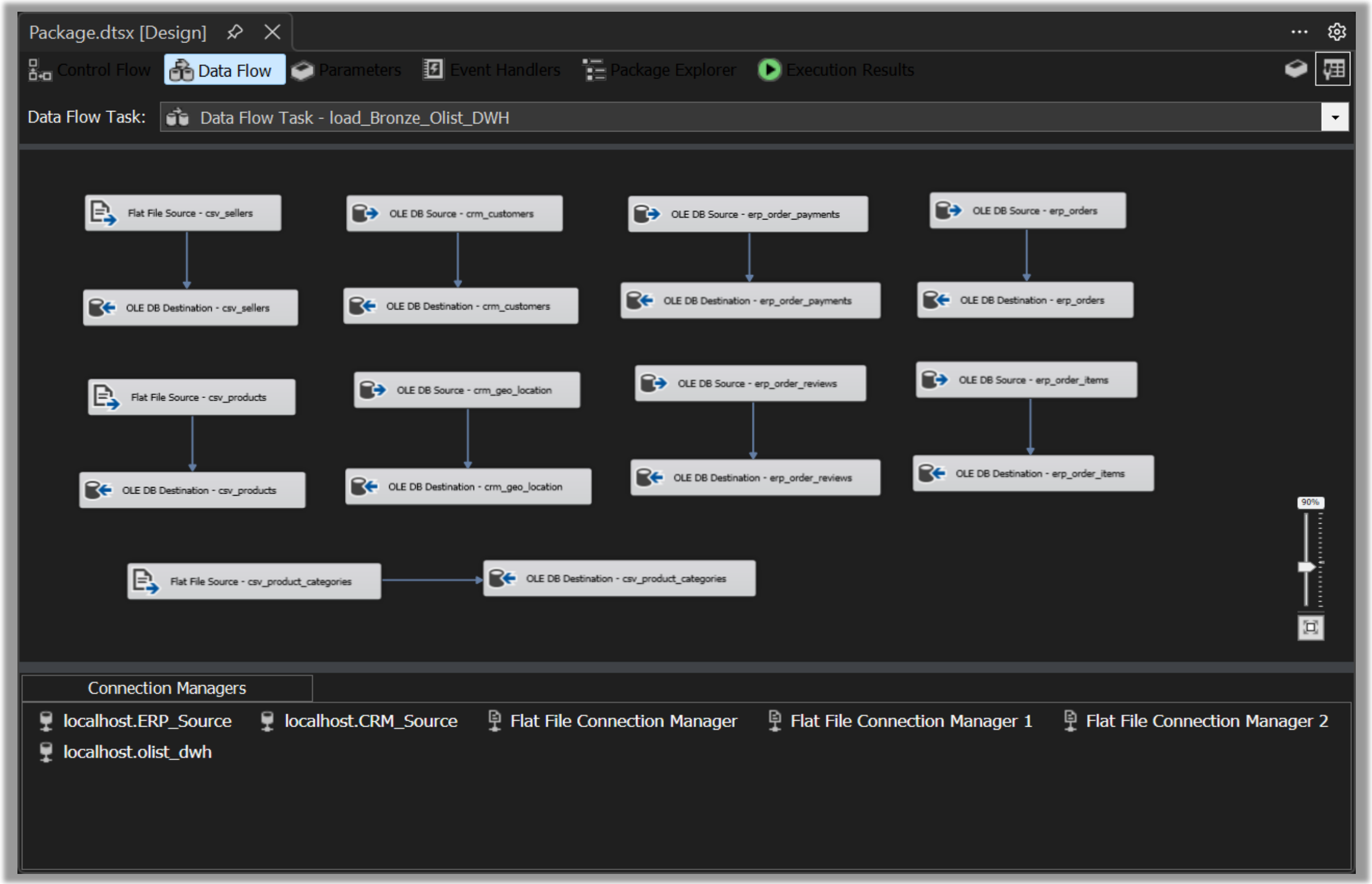
08 **High-Value Categories Require a Different Strategy**

The **Computers & Accessories** category recorded an average order value of approximately **R$1,286.18 across 181 orders**. This illustrates how some categories can generate high-value orders despite lower order volumes, supporting the use of category-specific sales strategies.

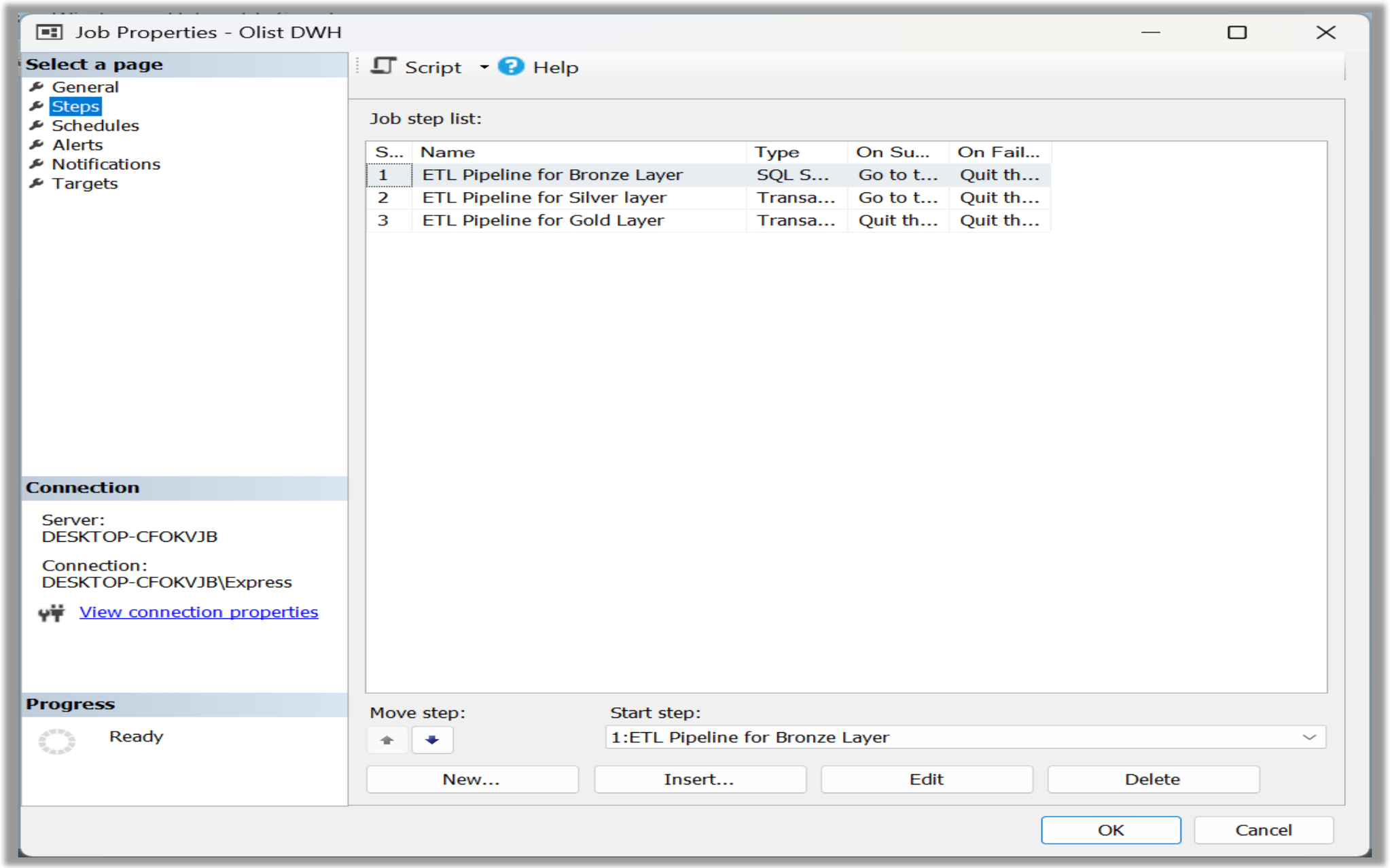
PHASE 03

# Documents

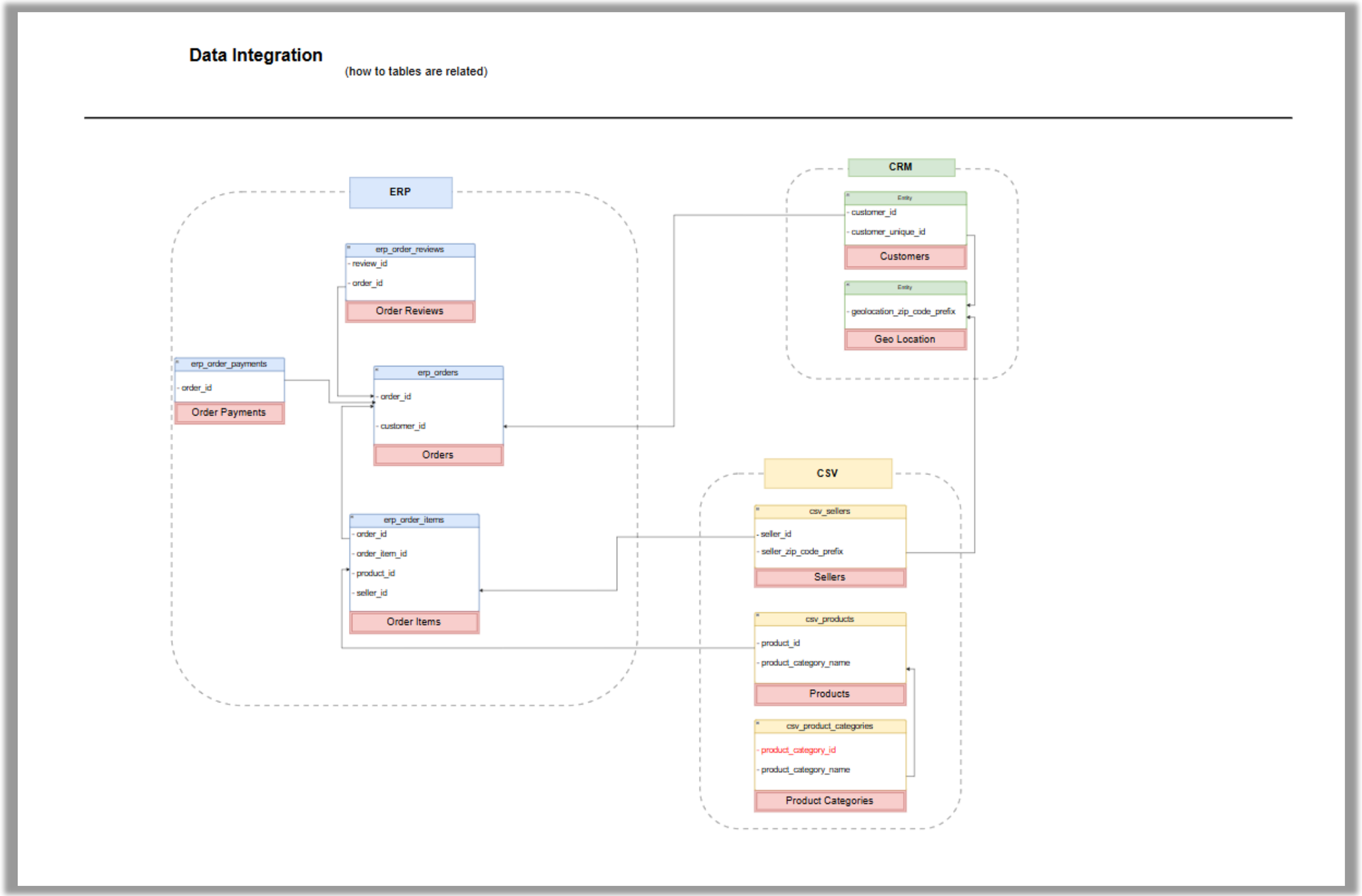
## SSIS ETL From Sources To Bronze Layer



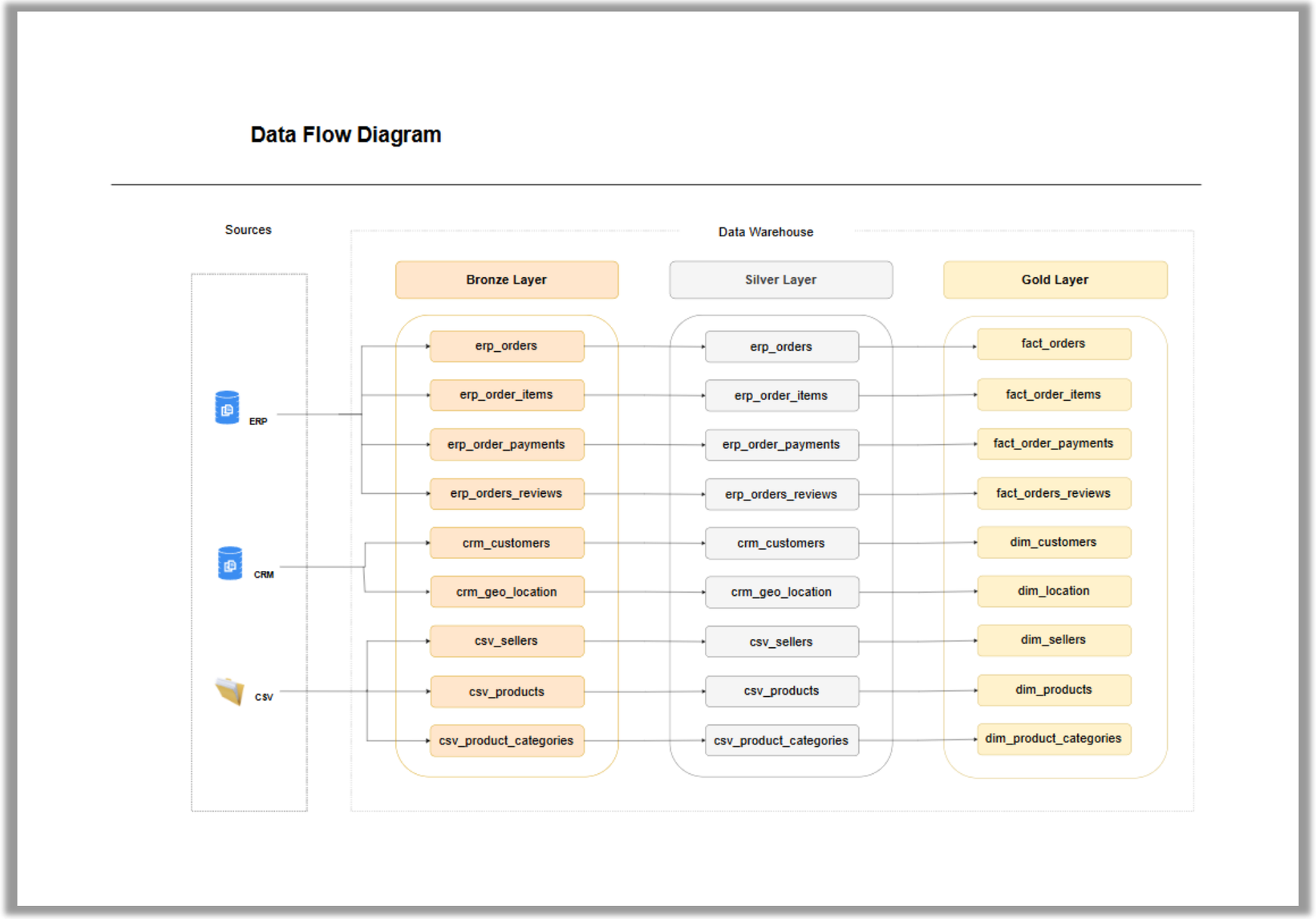
## ETL Steps in Job



## Data Integration Diagram

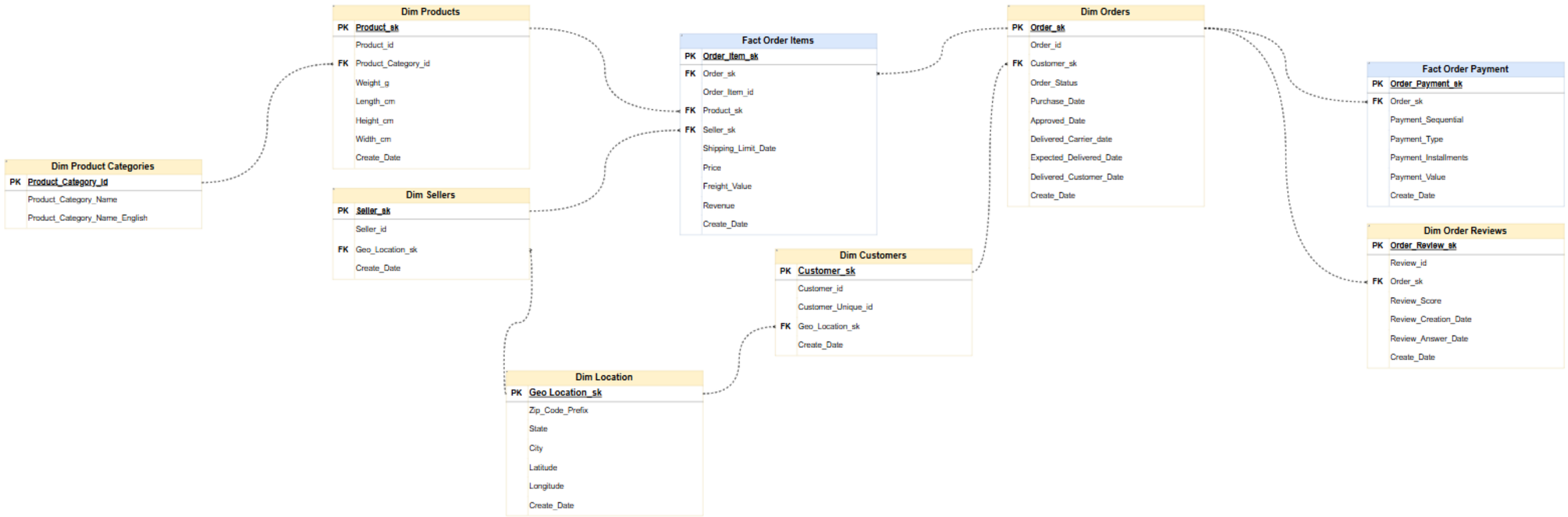


## Data Flow Diagram



## Data Model Diagram (Galaxy Schema)

**Data Mart** ( Galaxy Schema)



olist



## What's Next? — Power BI Integration

Transitioning SQL findings into interactive dashboard analytics.




**Next Milestone:**

Publishing dynamic executive dashboard with slicers, drill-downs, and automated refresh.

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**GitHub**: https://github.com/Ahmed-Elsayad99/SQL-Data-Warehouse-Project.git